INCH POUND

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24 October 1994
SUPERSEDING
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MILITARY SPECIFICATION

SURVIVAL KIT, INDIVIDUAL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two types of survival kits.

* 1.2 Classification. The survival kits shall be of the following types, as specified (see 6.2):

Type I - ax (wood handle) and saw

Type II - deleted (see 6.5)

Type III - deleted (see 6.5)

Type IV - ax with guard and leather grip

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FQSD, 2800 South 20th Street, Philadelphia, PA 19145-0599, by using Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8465

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2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and its supplement as cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-375	-Plastic Film, Flexible, Vinyl Chloride
V-F-106	-Fastener, Slide, Interlocking
V-L-61	-Laces, Footwear, Nylon
V-T-295	-Thread, Nylon
FF-N-845	-Nut, Plain, Wing
FF-R-556	-Rivets, Solid, Small; Rivets, Split Small; Rivets, Tubular, Small; and Caps, Rivets; General Purpose
FF-W-84	-Washers, Lock (Spring)
KK-L-165	-Leather, Cattlehide, Vegetable-Tanned and Chrome Retanned, Impregnated; and Soles
A-A-1037	-Wire Fabric, (Industrial)
TT-P-1757	-Primer Coating, Zinc Chromate, Low Moisture-Sensitivity
TT-L-32	-Lacquer, Cellulose Nitrate, Gloss, for Aircraft Use
CCC-C-419	-Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered
CCC-D-950	-Dyeing and Aftertreating Processes for Cotton Cloths
PPP-B-636	-Box, Fiberboard for Sealing and Securing
PPP-T-45	-Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

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MIL-B-131	-Barrier, Material, Water Vaporproof, Flexible
MIL-P-149	-Plastic Coating Compound, Strippable (Hot Dipping)
MIL-F-495	-Finish, Chemical Black, for Copper Alloys
MIL-W-530	-Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors
MIL-C-5040	-Cord, Nylon
MIL-I-6903	-Ink, Marking (For Parachutes And Other Textile Items)
MIL-C-8514	-Coating Compound, Metal Pretreatment, Resin-Acid
MIL-F-10884	-Fasteners, Snap
MIL-C-13924	-Coating, Oxide, Black, for Ferrous Metals
MIL-G-16491	-Grommet, Metallic
MIL-T-43566	-Tape, Textile, Cotton, General Purpose, Natural or in Colors

STANDARDS

FEDERAL

FED-STD-66	-Steel; Chemical Composition and Hardenability
FED-STD-151	-Metals; Test Methods
FED-STD-751	-Stitches, Seams, Stitchings

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MIL-STD-105	-Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-130	-Identification Marking of U.S. Military Property
MS27980	-Fasteners, Snap, Style 2 (Regular Wire Spring Clamp Type)
MS27982	-Fasteners, Snap, Style 3 (Pronged Ring Head Type)

(Unless otherwise indicated, copies of Federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government Publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

COLOR ASSOCIATION OF THE UNITED STATES, INCORPORATED

TCA Cable Numbers and Colors

(TCA cable numbers and colors are published by the Color Association of the United States, Incorporated, 200 Madison Avenue, New York, NY 10016-0927.)

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the survival tool kits furnished under this specification shall be products which have been inspected and have passed the first article inspection specified in 4.3 through 4.3.2.

3.2 Materials and components. The materials and components shall conform to the applicable specifications and standards as listed or required herein.

3.2.1 Blades

3.2.1.1 Ax. The ax blade shall be fabricated from straight, flat, through-hardened steel and shall be free from dirt, segregations, and inclusions. The thickness of the blade shall be 0.125 ± 0.010 inch when inspected as specified in 4.5.5. The blade shall have a Rockwell hardness of C47 ± 2 when determined 1/2 inch from the cutting edges as specified in 4.5.6. There shall be a hardness transition from the heel of the blade to the furthermost end of the tang. The hardness shall be less than that specified for the cutting edge, but shall be sufficient temper suitable for serviceability and peening of the end of the tang for the type IV kit ax.

3.2.1.2 Wood saw. The wood saw blade shall be fabricated from selected saw steel and shall be free from dirt, segregations, and inclusions. The thickness of the blade shall be 0.050 ± 0.005 inch when inspected as specified in 4.5.5. The wood saw blade shall be hardened uniformly throughout and shall have a Rockwell hardness of C47 ± 2 when determined at any point as specified in 4.5.6.

3.2.1.3 Hacksaw. The hacksaw blade shall be fabricated from steel suitable for the intended use. The thickness of the blade shall be 0.050 ± 0.005 inch when inspected as specified in 4.5.5. The teeth shall have a Rockwell hardness of C60 ± 2 when determined as specified in 4.5.6. The remainder of the blade shall not be hardened. The teeth shall be reground after setting and hardening to provide maximum clearance.

3.2.2 Scraper. The scraper shall be fabricated from through-hardened steel and shall be free from dirt, segregations, and inclusions. The thickness of the scraper shall conform to Figure 15 when inspected as specified in 4.5.5. The scraper shall have a Rockwell hardness of C43 ± 2 when determined as specified in 4.5.6.

3.2.3 Hacksaw blade tang. The hacksaw blade tang shall be fabricated from steel conforming to FED-STD-66, SAE 1010. The thickness of the hacksaw blade tang shall be 0.050 ± 0.005 inch when determined as specified on 4.5.5.

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3.2.4 Ax handle

3.2.4.1 Wood. The ax handle, for the type I kit ax, shall be fabricated from hard maple or birch wood. The handle shall be smoothly finished and shall be free of knots, knot holes, splits, cracks, decay, or other defects which may affect the appearance or serviceability.

3.2.4.2 Leather. The ax handle grip, for the type IV kit ax, shall be fabricated from the leather conforming to KK-L-165, type I or II, class 1, 2, 5 (subclass (d)), or 6, color natural as produced by the tanning process, cut into washers 3/16 inch thick. The leather shall be treated with the paranitrophenol fungicide.

3.2.5 Ax guard. The guard, for the type IV kit ax, shall be fabricated from cold rolled steel and shall conform to Figure 24.

3.2.6 Saw and scraper grip. The saw and scraper grip shall be fabricated from high-strength cast or forged commercial aluminum alloy.

3.2.7 Hook bolt. The hook bolt, used on the saw and scraper grip, shall be fabricated from stainless steel conforming to FED-STD-66, AISI 302.

3.2.8 Wing nut. The wing nut, used on the saw and scraper grip, shall be black oxide coated steel conforming to FF-N-845, type I, style A-1, size 8, 32UNC-2B threads per inch. As an alternate, the wing nut may be nickel plated in lieu of black oxide coating.

3.2.9 Lock washer. The steel lock washer, used on the hook bolt, shall conform to FF-W-84, class A, style 2, size No. 8, nominal size 0.164 inch. The lock washer shall be black oxide coated in accordance with MIL-C-13924, class 1. As an alternate, the lock washer may be nickel plated in lieu of black oxide coating.

3.2.10 Rivets

3.2.10.1 Ax handle. The rivets, for the type I kit ax handle, shall be fabricated from black oxide coated steel and shall have an oval head. The rivet shall be a two piece design and shall conform to Figure 7. The coating shall conform to MIL-C-13924, class 1.

3.2.10.2 Hacksaw blade. The oval head rivets, for attaching the hacksaw blade to the hacksaw blade tang, shall be steel conforming to FF-R-556, type XII, class 3, grade E, nominal size 1/8 inch. The length of the rivet shall be suitable for attaching the hacksaw blade to the hacksaw tang and shall not interfere with the insertion of the assembled hacksaw blade in the saw and scraper grip.

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* 3.2.11 Double-end hook. The double-end hook, for the ax carrying case, shall conform to Brookside Manufacturing Corporation Part Number 229 as manufactured by Brookside Manufacturing Corporation, 135 Wood Street, West Haven, CT 06516 or equivalent.

3.2.12 Grommets and washers. The grommets and washers shall be brass conforming to MIL-G-16491, type I, class 3, size number 0.

3.2.13 Reinforcing screen. The screen, for reinforcing the longitudinal inner edge of the carrying cases of the type I and IV ax kits, shall conform to A-A-1037, type I, class 1, grade D, phosphor bronze, 30 x 30 mesh.

3.2.14 Ferrule. The ferrule, used on type I kit ax handles, shall be fabricated from cold rolled steel and shall be seamless. The ferrule shall be tapered and shall conform to Figure 7. The ferrule shall be black oxide coated in accordance with MIL-C-13924, class 1.

3.2.15 Tape

3.2.15.1 Covering, spacer, and grommet tabs. The tape, for covering the reinforcing screen on the longitudinal inner edge of the ax carrying case and for the spacer and grommet tabs, shall be cotton conforming to MIL-T-43566, type I, class 4, one inch wide.

3.2.15.2 Thong. The slide fastener thongs, for the ax and saw carrying cases, shall be cotton tape conforming to MIL-T-43566, type I, class 4, 3/8 inch wide. The thong shall be looped through and securely knotted to the pull, approximately at the center of the thong length. The unattached ends of the thong shall be knotted together so that the knot is approximately one inch from the end. The finished length of the thong shall be 4 $\pm$ 1/4 inches.

3.2.16 Webbing

3.2.16.1 Hand Sling. The webbing, for the type I ax, saw and scraper grip hand slings, shall be cotton conforming to MIL-W-530, type IIa, class 4, 1/2 inch wide.

3.2.16.2 Neck band. The webbing, for the neck band for the ax carrying case, shall be cotton conforming to MIL-W-530, type IIa, class 4, one inch wide.

3.2.16.3 Hanger and saw carrying case pocket. The webbing, for the hanger of the ax carrying case and for the saw carrying case pocket, of the type I kit, shall be cotton conforming to MIL-W-530, type IIa, class 4, 2 inches wide.

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3.2.16.4 Belt loop. The webbing, for the type I saw kit carrying case belt loop, shall be cotton conforming to MIL-W-530, type IIb, class 4, 3 inches wide.

3.2.17 Slide fastener. The slide fastener, for the ax and saw carrying cases, shall conform to V-F-106, type I, style 3, size M, with a long pull tab. The chain and all the other metal components shall be brass with a black chemical finish. The slide fastener tape and bead shall be nylon, dyed to approximately match the color of the cloth of the ax and saw carrying cases (see 3.8.1), as applicable, and shall have at least "good" colorfastness to laundering and light. The opening in the pull tab shall be of sufficient size to accommodate the thong (see 3.2.15.2). The slide fastener length shall be governed by the flat pattern, Figure 6, 13, or 20, and Figures 1, 11, or 17, as applicable.

3.2.18 Snap fasteners

3.2.18.1 Ax and saw carrying case. The snap fasteners, for the ax and saw carrying cases shall conform to MIL-F-10884, style 2, finish 2, and MS 27980 -2B, -6B, -7B, and -8B.

3.2.18.2 Sharpening stone carrying case. The snap fasteners, for sharpening stone carrying case, shall conform to MIL-F-10884, style 3 or style 5, finish 2, and MS27982-1B, -3B, 4B, and -5B. As an alternative, similar commercial kinds of snap fasteners may be substituted.

3.2.19 Carrying case

3.2.19.1 Ax and saw. The carrying case, for the ax and saw kits, shall be fabricated from the cotton duck cloth conforming to CCC-C-419, type I, number 10 and shall be vat dyed (see 3.8.1), water repellent, and mildew-resistant aftertreated in accordance with CCC-D-950, type I, class B.

3.2.19.2 Sharpening stone. The carrying case for the sharpening stone shall be fabricated from the vinyl plastic film conforming to L-P-375, type II, class 2, 0.0080 inch thick.

3.2.20 Thread. The thread used in all the sewing operations shall be nylon conforming to V-T-295, type II, size B. The thread shall approximately match the color of the ax and saw carrying cases (see 3.8.1) and shall have at least "good" color fastness to light and laundering.

3.2.21 Cord. The cord, used as the flexible handle of the sharpening stone, shall be nylon conforming to MIL-C-5040, type 1A.

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3.2.22 Cord stop. The cord stop, used on the flexible handle of the sharpening stone, shall be fabricated from brass and shall conform to Figure 10. The cord stop shall be black chemically finished in accordance with MIL-F-495.

3.2.23 Sharpening stone. The sharpening stone, for the type I and IV ax kits, shall be made of suitably bonded aluminum oxide. One side shall be number 90 grain size, suitable for sharpening the various cutting tool edges. The other side shall be a fine grain, suitable for honing the various cutting tool edges.

3.2.24 Marking ink. The marking ink shall conform to MIL-I-6903, Color black and shall have a pH of 5.0 to 8.5.

3.2.25 Leg strap. The leg strap, for the type I and IV ax kits, shall be the nylon lace conforming to V-L-61, type I, class 1, 40 inches minimum in length.

3.2.26 Fire starter. The fire starter for type I and IV ax kits shall consist of a synthetic flint, a plastic cap on one end, and a flexible retaining loop. The flint shall be fabricated from a pyrophoric alloy substance adequately compounded for strength to withstand breaking and produce sparks of sufficient intensity to induce fire. The flint size shall be approximately 2-1/2 inches in length by 1/4 inch diameter. One end shall be securely embedded into a suitable compounded colored plastic cap approximately 15/16 inch in length by 3/8 diameter. The upper end of the cap shall contain a through hole for accommodating a flexible retaining loop. The retaining loop shall be constructed from a material resistant to corrosion, and shall measure, when extended, a minimum of 4 inches. The retaining loop, with an instruction card attached to it, shall be threaded through the hole in the cap and the ends joined securely together.

3.2.26.1 Coating and packaging of fire starter. Each fire starter shall be dipped in coating material conforming to MIL-P-149 type II, color light blue. The coating material shall cover the full length of the fire starter including a portion of the cap. The fire starter, a dessicant capsule 1/4" by 1-3/4", and instruction card shall be heat-sealed together within an envelope made of waterproof barrier material conforming to MIL-B-131, class 1.

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3.2.26.1.1 Instruction card. The instruction card shall contain the following information:

FIRE STARTER

IMPORTANT: Do not stow fire starter after removal from package.

USE INSTRUCTIONS

1. READY. Gather tinder (dry wood shavings, paper, grass, leaves, cloth, bark) and prepare other campfire material in the usual manner.

2. AIM. Use a sharp instrument (knife, sharp rock, glass) to slowly carve small particles of fire starter into concentrated area on tinder.

3. FIRE: Scrape the fire starter fast,.....throwing sparks at concentrated particles. The particles will ignite and start the tinder burning.

3.3 Design

3.3.1 Type I. The type I kit shall consist of the ax kit and saw kit. The two kits shall be joined together by snap fasteners and each kit shall contain specified tools.

* 3.3.1.1 Ax kit. The type I ax kit shall consist of the ax (wooden handle), the sharpening stone, the fire starter, the leg strap, and the ax carrying case. The ax carrying case shall have a slide fastener closure, two inner pockets (one on each side), a snap fastener neck band for holding the ax firm in case, and a webbing hanger with the double-end hook at the top for attaching to the pistol belt and to the grommets in the belt loop webbing. Each lower corner of the carrying case shall contain a grommet tab and the space between the tabs shall be filled with spacer tape. Three snap fastener studs shall be located on the underside of the ax carrying case and shall mate with the snap fastener sockets on the upper side of the saw carrying case. The sharpening stone shall be inserted into its carrying case and the carrying case shall be closed. The sharpening stone, in its carrying case, and the fire starter shall be stowed individually in the pockets of the ax carrying case. The leg strap shall also be stowed in one of the pockets. The ax carrying case, when closed with slide fastener and with the neck band snapped closed, shall provide a snug fit, allowing a minimum of movement of the contents.

* 3.3.1.2 Saw kit. The saw kit shall consist of a saw and scraper grip with the internal snow-saw teeth, the wood and

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hacksaw blades, the scraper, and the saw carrying case. The saw carrying case shall have a slide fastener closure and one inner diagonal pocket fabricated from webbing. Three snap fastener sockets shall be located on the upper side of the saw carrying case and shall mate with the three snap fastener studs on the underside of the ax carrying case. The belt loop shall be attached to the top of the saw carrying case and shall contain two grommets for attaching to the double-end hook of the previous survival kit. When closed, the carrying case shall provide a snug fit, allowing minimum movement of the contents.

3.3.3 Type IV. The type IV kit shall consist of the ax with the guard and leather grip, the sharpening stone, the fire starter, the leg strap, and the ax carrying case. The ax carrying case shall have a slide fastener closure, two inner pockets (one on each side), the snap fastener neck band for holding ax firm in the case, and the webbing hanger with the double-end hook at the top for attaching to the pistol belt. Each lower corner of the carrying case shall contain a grommet tab and the space between the tabs shall be filled with the spacer tape. The sharpening stone shall be inserted into its carrying case and the carrying shall be snapped closed. The sharpening stone, in its carrying case, and the fire starter shall be stowed individually in the pockets of the ax carrying case. The leg strap shall also be stowed in one of the pockets. The ax carrying case, when closed with the slide fastener and with the neck band snapped closed, shall provide a snug fit, allowing a minimum of movement of the contents.

3.4 Construction

3.4.1 Seams and stitching. All the seams used in the fabrication of the ax and saw carrying cases shall conform to Figures 1 through 5, 11, 12, and 17 through 19, as applicable, and shall be accomplished with Stitch Type 301, conforming to FED-STD-751, with 8 to 10 stitches per inch. Except for the thong and sling, no cut or raw edges shall be visible on the exterior of the carrying case. All the seam edges shall be properly forced out and shall not contain any folds. No seam shall be twisted, puckered, or pleated and no part of the carrying case shall be caught in an unrelated operation or seam. All the thread ends shall be trimmed to length of 1/4 to 3/8 inch.

3.4.2 Sewing. Each row of stitching shall be straight and parallel to the seam edge. The straightness of the stitching in any row shall be maintained within a tolerance of plus or minus 1/32 inch. The ends of the stitching, that are not caught in other seams or stitching, shall be securely backstitched, by overlapping on itself, for at least 1/2 inch. Thread breaks, skips, and run-offs shall be overstitched not less than one inch. The thread tension shall be maintained so that there shall not be any loose or tight stitching and the lock shall be embedded in the materials sewn together.

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3.4.3 Construction details

3.4.3.1 Type I

3.4.3.1.1 Ax carrying case. The ax carrying case and pockets shall be fabricated in accordance with Figures 1 through 6. The slide fastener shall close upward and shall form the forward edge of the case when it is worn on the right hip.

3.4.3.1.2 Ax. The ax blade shall conform to Figure 8. The cutting sides of the ax blade and the sickle hook shall be sharpened to a keen edge. The sickle hook edge shall be one-bevel, flat ground, recessed and curved as specified in Figure 8. The ax handle shall be a one-piece construction and suitably grooved to receive the tang of the ax. The tang of the blade shall be securely cemented within the groove of the handle, and no portion of the tang shall protrude beyond the edges of the handle. The ferrule shall be beveled and shall be a snug fit to the handle. The rivets shall be inserted through the handle and blade and through the ferrule, handle, and blade as specified in Figure 7. The rivet heads shall be driven tight against the handle and the ferrule, as applicable. One end of the handsling shall be threaded through the top hole in the handle and the ends shall be knotted together so that the knot is approximately one inch from the cut ends. The assembled ax shall conform to Figure 7.

3.4.3.1.3 Sharpening stone carrying case. The sharpening stone carrying case shall be fabricated in accordance with Figure 9. The sharpening stone shall fit properly in the carrying case.

3.4.3.1.4 Sharpening stone. The sharpening stone shall be constructed solidly of two grain sections as specified in Figure 10 (see 6.3). The coarse and fine grain sections shall be fused together after forming and shall result in one integral unit. The color of the coarse grain side shall be a dark color and the color of the fine grain side shall be a light contrasting color. The sharpening stone shall have rounded or 90 degree edges and shall have a center opening suitable for accommodating the nylon cord. A doubled, single, continuous length of the nylon cord, without any splices or knots except at the ends, shall be inserted through the center opening in the sharpening stone and the cord stops shall be attached to the cord as specified in Figure 10. Each end of the doubled cord shall be knotted together, and the cord stops shall be crimped in a manner so that they shall not pass through the center opening of the sharpening stone. The crimped flared end of the stop, when positioned in the center opening of the sharpening stone, shall be flush or just below the surface of the stone. The cord stops shall be crimped without distortion, damage, splitting, or cutting of the cord. The center opening of the sharpening stone shall not restrict the movement of the cord. The sharpening stone shall be inserted into its carrying case prior to stowing in one of the pockets of the ax carrying case.

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3.4.3.1.5 Fire starter. The fire starter shall be stowed in the pocket of the ax carrying case that does not contain the sharpening stone.

3.4.3.1.6 Leg strap. The leg strap shall be neatly folded and shall be inserted into one of the pockets of the ax carrying case.

3.4.3.2 Saw carrying case. The saw carrying case and pocket shall be fabricated in accordance with Figures 11 through 13. The slide fastener chain shall be located on the front saw carrying case, as specified in Figure 11, when the case is worn on the right hip. The slide fastener shall close upward.

3.4.3.2.1 Saw and scraper grip. The saw and scraper grip shall be of a one-piece construction and shall contain eight snow-teeth. The edges of the snow-saw teeth shall lie in the same plane and shall be sufficiently sharp for removal of packed snow when used in conjunction with the wood saw as a snow block cutting tool. At the blade attachment end, the saw and scraper grip shall have four blade alignment projections, the hook bolt, the lock washer, and the wing nut. The extreme threaded end of the hook bolt shall be flattened on one side, for 1/8 inch, to prevent the removal of the wing nut. One end of the handsling shall be threaded through the hole in the handle and the ends shall be knotted together so that the knot is approximately one inch from the cut ends. The saw and scraper grip shall conform to Figures 14 and 15 (see 6.3).

3.4.3.2.2 Wood saw blade. The wood saw blade shall have every tooth set the same amount on each side of the blade with the alternate teeth being set to the same side of the blade. The teeth shall be sufficiently sharp for cutting either wood logs, packed snow, or ice. The back edge, near the tip, shall be sharpened to keen edge so that it can be used for skinning or as an emergency knife. The tip end shall have an ice pick edge with a beveled point. The wood saw blade shall conform to Figure 16.

3.4.3.2.3 Hacksaw blade. The hacksaw blade shall be securely attached to the hacksaw blade tang by the oval head rivets. The rivets shall not interfere with the attaching of the assembled hacksaw blade to the saw and scraper grip. The assembled hacksaw blade shall conform to Figure 16 (see 6.3).

3.4.3.2.4 Scraper. The scraper shall have a beveled working edge and shall conform to Figure 15.

3.4.3.2.5 Interchangeability of the tools. The wood saw blade, the hacksaw blade, and the scraper shall have attaching ends of identical size and shape and with identical elongated holes for the hook bolt connection. The shoulders on the attaching ends of

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the saw blades and the scraper shall be located at identical distances, so that the blades and the scraper are interchangeable in the saw and scraper grip. The blades and scraper shall fit snug and lay flat when attached to the saw and scraper grip.

3.4.3.3 Type IV

3.4.3.3.1 Ax carrying case. The ax carrying case and pockets shall be fabricated in accordance with Figures 2, 3, and 17 through 20. The slide fastener shall close upward and shall form the forward edge of the case when it is worn on the right hip.

3.4.3.3.2 Ax. The ax blade shall conform to Figures 22 and 23. The cutting sides of the ax blade and the sickle hook shall be sharpened to a keen edge. The sickle hook edge shall be one-bevel, flat ground, recessed and curved as specified in Figure 23. The trenching edge shall be beveled but need not be sharpened. The guard shall conform to Figure 24 and shall be inserted on the tang as specified in Figure 21. The tang shall fit into the guard so that all the edges of the bottom slot are flush on all the sides about the tang. The sides of the tang, which are near the bottom slot are flush on all sides about the tang. The sides of the tang, which are near the bottom slot of the guard, shall be peened against the guard as specified in Figure 21. Sufficient leather washers shall be inserted on the tang to cover the entire tang between the ends of the guard and to form the grip. The washers shall fit snugly on the tang. The top of the guard shall be inserted on the tang. The top end of the tang shall protrude through the top end of the guard. The top of the tang shall be upset to fill the recess in the top of the guard and shall form a flat or rectangular head on the guard. All sharp edges shall be removed. The upset portion of the tang shall not be ground flush with the surface of the guard. The upsetting of the tang shall also firmly compress together the leather washers on the guard. The washers shall follow the contour of the tang. The washers shall blend and shall be uniform in height and width and shall be smooth in appearance. The sides of the leather washers, at the bottom of the guard, shall be flattened as specified in Figure 21. All the edges of the guard shall be deburred to eliminate sharp edges. The assembled ax shall conform to Figure 21.

3.4.3.3.3 Sharpening stone case. The sharpening stone carrying case shall be fabricated in accordance with Figure 9. The sharpening stone shall fit properly in the carrying case.

3.4.3.3.4 Sharpening stone. The sharpening stone shall be the same as specified in 3.4.3.1.4.

3.4.3.3.5 Fire starter. The fire starter shall be the same as specified in 3.4.3.1.5.

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3.4.3.3.6 Leg strap. The leg strap shall be the same as specified in 3.4.3.1.6.

3.5 Assembling of the components. The components shall fit properly in their respective carrying cases. For the type I kit only, the ax carrying case shall be capable of being snapped to the saw carrying case without distortion.

3.6. Weight. When weighed as specified in 4.5.2, the weight of the assembled tool kits shall not exceed the following:

TYPE	Maximum weight, pounds
I	Ax kit, 2.0; Saw kit, 1.25
IV	2.0

3.7 Performance inspections

3.7.1 Deformity of the blades (ax, wood saw, and hacksaw) and the scraper. Any concave or convex deviation of the ax, wood saw, and hacksaw blades and the scraper, as applicable, shall not exceed 1/16 inch, when inspected as specified in 4.5.3.

3.7.2 Saw cutting.

* 3.7.2.1 Wood saw blade. When inspected as specified in 4.5.8.1, the wood saw blade shall be capable of making a minimum of four - one inch deep cuts without the breaking of the blade or teeth or any damage to the teeth or saw and scraper grip. The time required to make each one inch deep cut shall not exceed 5 minutes.

3.7.2.2 Hacksaw blade. When inspected as specified in 4.5.8.2, the hacksaw blade shall be capable of cutting the steel tube into eight sections without the breaking of the blade or teeth or any damage to the teeth or the saw and scraper grip. No rivet shall become loose or break.

3.7.3 Stress of the scraper. When inspected as specified in 4.5.9, the scraper shall not break, deform, or crack.

3.7.4 Sharpening of the ax blade. When inspected as specified in 4.5.10, the sharpening stone shall be capable of sharpening and honing the ax blade and sickle hook cutting edges without disintegrating or damaging the ax blade.

* 3.7.5 Ax cutting. When tested as specified in 4.5.7, the ax shall be capable of cutting through vines, shrubbery, tree sprouts, saplings, etc. without damage to the ax.

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3.8 Color

3.8.1 Ax and saw carrying cases. The color of the cotton cloth for the ax and saw carrying cases shall be Olive Drab and approximately match TCA Cable No. 66022, Shade S-1 (U.S. Army Olive Drab).

3.8.2 Tape, webbing, cord, and leg strap. The color of the tape, webbing, cord, and leg strap shall approximately match the color of the cotton cloth used for the ax and saw carrying cases (see 3.8.1).

3.8.3 Sharpening stone carrying. The color of the sharpening stone carrying case shall be bright red.

3.8.4 Ax handle and saw and scraper grip. The ax handle and the saw and scraper grip of the type I kit shall be finished in a bright red color. The finish shall not crack, chip, or scale in normal usage.

3.9 Finish

3.9.1 Ax guard. The type IV ax guard shall be uniformly coated, first with the metal pretreatment coating compound conforming to MIL-C-8514, then with the zinc chromate primer conforming to TT-P-1757, color Y or T, and finally with the cellulose nitrate lacquer conforming to TT-L-32, color gloss black. Prior to the insertion of the leather washers, the upset portion of the side edges of the ax blade tang at the bottom slot of the guard shall be coated with the black cellulose nitrate lacquer. After the top of the tang has been upset on the top of the guard surface, the upset surface shall be uniformly coated, first with the metal pretreatment coating compound, then with zinc chromate primer, and finally with the black cellulose nitrate lacquer.

3.9.2 Ax, wood saw, and hacksaw blades and the scraper. The entire surfaces of the ax, wood saw, and hacksaw blades and the scraper, as applicable, shall be uniformly coated with the black oxide coating compound conforming to MIL-C-13924, class 1 (see 4.5.4). After all the specified edges of the ax and saw blades and the scraper, as applicable, have been sharpened, the corrosion preventative compound Rust Proof X-700 as manufactured by E.G. Walters, Inc., Orwigsburg, Pennsylvania, or equivalent, shall be uniformly applied to all the exposed metal surfaces. After the application of the corrosion preventative compound, the forward portion of the wood saw blade shall be encased in a snug fitting bag fabricated from the vinyl plastic film. The bag shall be of sufficient length to fully cover the entire sharpened back edge and shall be of a reasonably snug fit.

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3.10 Markings. All the identification markings shall be in accordance with MIL-STD-130 and shall be not less than 1/4 inch high. All markings shall be thoroughly dry prior to packing or packaging.

* 3.10.1 Ax and saw carrying cases. The underside of each ax and saw carrying case, as applicable, shall be marked with the marking ink, as follows:

SURVIVAL KITS; INDIVIDUAL, TOOL
NAME OF MANUFACTURER
TYPE (AS APPLICABLE)
MIL-S-8642F
CONTRACT NO.
NSN

The carrying case for the type IV kit shall also contain the 1/2 inch high markings specified in Figure 17.

* 3.10.2 Tools. Each tool shall be marked, in accordance with MIL-STD-130, as follows:

NAME OF MANUFACTURER OR TRADE MARK
NAME OF TOOL
MIL-S-8642F

* 3.11 Workmanship. After completion of the final assembly, the survival tool kits shall be thoroughly cleaned and all loose thread, lint, and foreign matter shall be removed. Each grommet and snap fastener shall be clinched without distortion, damage, splitting, or cutting of the cloth, webbing, or tape, as applicable. The tool kits shall not contain any crack, nick, burr, unspecified sharp edge or hole, tear, needle chew, mend, patch, spot, stain corrosion or scale, pit, or dent. Because of the emergency use of these items, the importance of providing a product of uniform excellent quality cannot be overemphasized. The tool kits shall be uniform in quality and shall be free from irregularities or defects which could adversely affect performance, reliability, or durability. The survival tool kits shall conform to quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the kits shall be classified as follows:

a. First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.3 through 4.3.2).

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.4 through 4.4.1.2).

4.3 First article inspection. The first article inspection of the kits shall consist of examination and tests for all the requirements of this specification.

4.3.1 First article samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit five kits of each type specified in the contract or order. The samples shall be representative of construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these kits from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of procuring activity (see 6.2). Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the Contracting Officer (see 6.2.).

4.3.2 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. One of the approved first article inspection sample kits of each type, as applicable, will be returned to the manufacturer for use in monitoring production. The other four kits of each type, as applicable, wil be destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract.

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4.4 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

Visual examination of the kits
Dimensional check of the kits
Weight of the kits
Deformity of the ax, wood saw, and hacksaw blades and the scraper, as applicable
Finish of the ax, wood saw, and hacksaw blades and scraper, as applicable
Thickness of the ax, wood saw, and hacksaw blades, hacksaw blade tang, and the scraper, as applicable
Rockwell hardness of the ax, wood saw, and hacksaw blades and the scraper, as applicable
Cutting the ax, wood saw, and hacksaw blades, as applicable
Stress of the scraper Sharpening of the ax blade Packaging

4.4.1 Sampling

4.4.1.1. Inspection lot

4.4.1.1.1. Kits. An inspection lot size shall be expressed in units of one kit of one type made essentially under the same conditions and from the same materials and components. The sample unit shall be one kit of one type.

4.4.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing kits of one type, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing kits of one type, fully prepared for delivery with the exception that it need not be sealed.

* 4.4.1.2 Sampling for the examinations and tests of the kits. The sample size, acceptance criteria, examinations, and tests required for the kits shall be as specified in MIL-STD-105.

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TABLE I.
Sample size

Test and examinations of the kits

Inspection	Paragraph	
	Requirement	Method
Visual examination	3.4 through 3.4.3.3.6, 3.5, 3.8 through 3.8.4, 3.9 through 3.9.2, 3.10 through 3.10.2 3.11 and Figures 1 through 25, as applicable	4.5.1.1 and Table II
Dimensional check	3.2.26 and Figures 1 through 25, as applicable	4.5.1.1 and Table III
Weight	3.6	4.5.2
Deformity of the ax, wood saw, and hacksaw blades and the scraper, as applicable	3.7.1	4.5.3
Finish of the ax, wood saw, and hacksaw blades and the scraper, as applicable 2/3/5/6/	3.9.2	4.5.4

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TABLE I.
Sample size

Test and examinations of the kits (Continued)

Inspection	Paragraph	
	Requirement	Method
Thickness of the ax, wood saw,and hack -saw blades, hacksaw blade tang,and the scraper as applicable	3.2.1 through 3.2.1.3, 3.2.2, and 3.2.3 as applicable	4.5.5
Rockwell hardness of the ax; wood saw, and hacksaw blades and the scraper as applic-able 3/ 5/	3.2.1 through 3.2.1.3, and 3.2.2 as applicable	4.5.6
Cutting of the ax, wood saw, and hacksaw blades, as applicable	3.7.2 through 3.7.2.2 and 3.7.5, as applicable	4.5.7 and 4.5.8 through 4.5.8.2 as applic-able
Stress of scraper 4/	3.7.4	4.5.9
Sharpening of the ax blade	3.7.5	4.5.10
Packaging	Section 5	4.5.1.2

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1/ The sampling plan acceptance numbers shall apply collectively to all the characteristics within a stated acceptable quality level.

2/ The thickness and hardness of the ax, wood saw, and hacksaw blades and the scraper, as applicable, shall be determined on every ax, wood saw, and hacksaw blade and scraper used in the finish inspection.
The finish of the ax, wood saw, and hacksaw blades and scraper, as applicable, shall be determined first and then the blades and scraper, as applicable, shall be inspected for thickness and hardness.

3/ The cutting of the ax, wood saw, and hacksaw blades, as applicable, shall be determined on every ax, wood saw, and hacksaw blade used in the hardness inspection. The hardness of the ax, wood saw, and hacksaw blades shall be determined first and then the blades, as applicable, shall be inspected for cutting.

4/ This inspection is only applicable when the type I kit is purchased.

5/ The stress of the scraper shall be determined on every scraper used in the hardness inspection. The hardness of the scraper shall be determined first and then the scraper shall be inspected for stress.

6/ The sharpening stone shall be obtained from every type I or IV kit, as applicable, used in the cutting inspection and shall be used to sharpen the axes used in the cutting inspection. Use one sharpening stone for each ax. The cutting inspection of the ax blades shall be determined first, and then the ax blades shall be used in the sharpening inspection.

4.5 Inspection methods

4.5.1 Visual examination

4.5.1.1 Kits. The contents and the carrying case of each of the kits of one type, selected as a sample unit from the lot, shall be thoroughly checked dimensionally and examined visually, as applicable, to determine conformance to this specification. The classification and list of defects, Tables II and III, as applicable, shall be used to classify and enumerate the defects found.

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TABLE II.

Classification of defects for the visual examination of the kits

Defect	Major	Minor
MATERIAL DEFECTS AND WORKMANSHIP DAMAGES		
GENERAL		
a. Any non-specified hole, scissors or knife cut, tear burn, or weakening defect such as a smash, multiple float, loose slub, needle chew, or abraded area	1/	
b. Color of any component not as specified or contains one or more shade bars		X
c. Any thread end not trimmed to a length of 1/4 to 3/8 inch or loose thread ends not removed		X
d. Any hand sling not knotted	X	
e. Any spot or stain	1/	
METALLIC COMPONENTS		
a. Any surface rough or contains any crack, nick, burr, unspecified sharp edge, dent, or sliver	X	
b. Any surface unclean or contains embedded foreign matter		X
c. Any component loose, detached, or otherwise not securely retained	X	
d. Any component malformed, misaligned, corroded, fractured, broken, chipped, bent, or distorted	X	
e. Finish or preservative compound not uniformly applied to any tool, as applicable	1/	

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TABLE II.
(Continued)

Classification of defects for the visual examination of the kits

Defect	Major	Minor
f. Any cord stop snap fastener or grommet loose, improperly finished, or improperly clinched resulting in cutting of the cord cloth, tape, or webbing, as applicable	X	
g. Any snap fastener mismated or misplaced 2/	X	
h. Any part of the slide fastener assembly bent, broken, or movement of the slider interfered with 3/	X	
i. Any finish or preservative compound not as specified	1/	
j. Any blistering, peeling, or flaking of the finish or preservative compound		X
COMPONENTS AND ASSEMBLY		
a. Any component not as specified or any defect of a component or defect of assembly not herein classified	1/	
b. Any component, component part, or required operation omitted, or any operation improperly performed not herein classified	1/	
SEAMS AND STITCHING		
a. Accuracy of Seaming:		
1. Any seam, hem, or attachment of any component twisted, puckered, pleated or caught in an unrelated operation or stitching; any seam edge not properly forced out or contains any fold		X

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TABLE II.
(Continued)

Classification of defects for the visual examination of the kits

Defect	Major	Minor
2. Ends of stitching that are not caught in other seams or stitching:		
(a) Not securely backstitched	X	
(b) Secure but backstitched for a distance less than 1/2 inch or the backstitching is not overlapped on itself		X
3. Thread breaks:		
(a) Not securely overstitched	X	
(b) Secure but overstitched for a distance less than one inch		X
b. Gage of stitching 4/:		
1. Any stitching irregular, unevenly gaged, or various stitchings not uniform		X
c. Open seams 5/:		
1. One inch or less		X
2. More than one inch	X	
d. Raw edges 6/:		
1. More than 1/4 inch but not more than one inch		X
2. More than one inch	X	
e. Seam and stitch type:		
1. Not specified seam or stitch type	X	

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TABLE II.
(Continued)

Classification of defects for the visual examination of the kits

Defect	Major	Minor
f. Stitch tension 7/:		
1. Loose tension, resulting in a loose seam, or tight tension, resulting in breaking of stitches, when normal pull is applied	X	
g. Stitches per inch 8/:		
1. Stitches more or less than specified:		
(a) One or two		X
(b) More than two	X	
MARKINGS		
a. Type missing, incomplete, incorrect, or illegible	X	
b. Other than type information missing, incomplete, incorrect, or illegible		X
c. Improperly located or size of markings not as specified	X	
AX		
a. Handle :		
1. Finish chipped, cracked, or scaled		X
2. Contains any knot, split, crack, or fabricated from more than one piece	X	

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TABLE II.
(Continued)

Classification of defects for the visual examination of the kits

Defect	Major	Minor
3. Tang of the blade not flush with the sides of the handle	X	
4. Ferrule not seamless or beveled	X	
5. Any rivet not driven flush to the handle or ferrule or not a two piece design	X	
6. Leather washers not firmly compressed together, not uniform in height and width, or the sides at the bottom of the guard are not flattened and tapered as specified 9/	X	
b. Blade:		
1. Cutting sides not convex ground, radial curved or sharpened to a keen edge	X	
2. Sickle hook not one bevel flat ground, recessed, curved, or sharpened to a keen edge	X	
3. Trench edge not beveled	X	
4. Tang not cemented within the groove of the handle	X	
5. Upset portion of the tang ground flush to the guard surface	X	
SHARPENING STONE:		
a. Not constructed as specified	X	
b. Cord contains any splice or knot, except at the ends	X	

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TABLE II.
(Continued)

Classification of defects for the visual examination of the kits

Defect	Major	Minor
c. Cracked, broken, chipped, or contains any sharp edge	X	
d. Carrying case not constructed as specified or sharpening stone does not fit properly		X
e. Sharpening stone not inserted into the carrying case, when placed in the pocket of the ax carrying case, or in the same pocket as the fire starter		X
f. Any end of the cord not knotted	X	
g. Any cord stop, when positioned in the center opening of the sharpening stone, not flush or just below the surface of the stone	X	
FIRE STARTER:		
a. Flint broken, cracked, malformed, misaligned, chipped, bent, distorted, or contains any sharp edge or dent	X	
b. Any component loose, detached, or not securely retained		X
c. Fire starter in the same ax carrying case pocket as the sharpening stone		X
AX CARRYING CASE:		
a. Pockets not top-open type or full use of any pocket not available	X	
b. Any snap fastener stud not aligned with the socket causing excessive bulge, when snapped to the saw carrying case, when applicable	X	

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TABLE II.
(Continued)
Classification of defects for the visual examination of the kits

Defect	Major	Minor
DEFECTS APPLICABLE TO THE TYPE I KIT ONLY		
SAW AND SCRAPER GRIP		
a. Finish chipped, cracked, or scaled	X	
b. Threaded end of the hook not flattened or flattened more than 1/4 inch	X	
c. Threaded end of the hook bolt flattened less than 1/16 inch or more than 3/16 inch		X
d. Blades or scraper do not fit snug or lie flat, when inserted in the saw and scraper grip (due to defect in the grip) 10/	X	
WOOD SAW BLADE		
a. Teeth not set the same amount or alternately	X	
b. Sharpened back edge not covered with the plastic bag or back edge not sharpened	X	
c. Does not fit snug or lie flat in the saw and scraper grip (due to defect in the blade) 10/	X	
HACKSAW BLADE		
a. Not reground after hardening	X	
b. Does not fit snug or lie flat in the saw and scraper grip (due to defect in the blade) 10/	X	

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TABLE II.
(Continued)

Classification of defects for the visual examination of the kits

Defect	Major	Minor
SCRAPER		
a. Working edge not beveled	X	
b. Does not fit snug or lie flat in the saw and scraper grip (due to defect in the scraper) 10/	X	
SAW AND CARRYING CASE		
a. Pocket not side open type or full use of pocket not available	X	
b. Female portion of the snap fastener not aligned with the studs causing excessive bulge when snapped to the Type I ax carrying case, when applicable	X	
c. Grommets in belt loop do not mate with the double-end hook of the Type I carrying case or the double-end hook cannot be inserted into the grommets due to defect in the grommets, improper location of grommets, or construction of the belt loop	X	

1/ The defect shall be considered major when it seriously affects the serviceability or appearance, otherwise it is to be classified as a minor defect.

2/ The snap fasteners shall be checked for proper function and attachment by snapping closed and unsnapping each of the snap fasteners at least three times.

3/ The slide fasteners shall be checked for proper function by opening and closing the slide fastener, at least three times, along its entire length.

4/ Gage of stitching defects are not to be scored, only when the condition exists on the major portion of the seam.

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5/ A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more continuous skipped stitches or run-offs occur. On double stitched seams, a seam is considered open when either one or both sides of the seam is open.

6/ Any edge that should be turned under and is not, but is securely caught in the stitching, shall be classified as a raw edge. Any raw edge not securely caught in the stitching shall be classified as an open seam.

7/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident, other than the gathering of the material, the seam hem, as applicable shall be tested by exerting normal pull in the lengthwise direction of the seam, hem, or stitching. Gathering of the material shall be recorded as puckered seam or hem, as applicable.

8/ Stitches per inch defects are to be scored only when condition exists on the major portion of the seam.

9/ These defects are applicable only to the type IV kit.

10/ Each blade and scraper shall be inserted and attached to each saw and scraper grip to determine whether the blades and scraper fit properly.

TABLE III.

List of defects for the dimensions of the assembled kits

Examine	Defect
Measure the carrying case, slide fastener, thong, neck band, double-end hook, grommet tabs, hanger, pockets, covering tape, sharpening stone, fire starter, ax, ferrule, rivets, hand sling, cord stop, leg strap, saw and scraper grip, hook bolt, lock washer, wood saw blade 1/, assembled hacksaw blade 1/, and scraper as applicable.	Any measurement deviating from the dimensions and tolerances specified in 3.2.26 and Figures 1 through 24, as applicable, shall be enumerated as a dimensional defect.

1/ The number of teeth per inch shall also be determined.

4.5.1.2 Packaging examination. Each of the fully prepared shipping containers containing assembled kits of one type, selected as a sample unit from the lot, shall be visually

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examined to determine that the packaging, packing and marking conform to this specification. The list of defects, Table IV, shall be used to enumerate the defects found.

TABLE IV. List of defects for packaging

Item	Defect
Exterior and interior markings	Missing, incomplete, incorrect, illegible; of improper size, location, sequence, or method of application; type on the kit not the same as the containers; markings not the same on the interior packages and exterior containers.
Materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Gross or net weight exceeds the requirement; more than one type kit in the same container

4.5.2 Weight. The weight of the assembled kit shall be determined on any apparatus capable of weighing to nearest 0.01 pound.

4.5.3 Deformity of the blades (ax, wood saw, and hacksaw) and the scraper. A straightedge shall be placed against the side of the blade at the sharpened edge of the ax blade and at the toothedge of the wood saw and hacksaw blades, as applicable, and then at the opposite edge. The straightedge shall be placed along the blade of the scraper and then along the tang portion. Any concave or convex deviation in the contacting blade edge or tang, as applicable, for the straightedge shall be determined by any suitable measuring device.

4.5.4 Finish of the blades (ax, wood saw, hacksaw, and the scraper). The acceptability of the black oxide coating on the blades (ax, wood saw, and hacksaw) and the scraper, as applicable, after the corrosion preventative compound has been removed, shall be determined in accordance with the procedure described in MIL-C-13924 for "Oxalic Acid Spot Test".

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4.5.5 Thickness of the blades (ax, wood saw, and hacksaw), hacksaw blade tang, and the scraper. The thickness of the blades, (ax, wood saw, and hacksaw), hacksaw blade tang, and the scraper, as applicable, shall be determined by any standard commercial method. The finish shall be removed prior to determining the thickness. Three thickness determinations shall be made on each blade portion of the tools and also on the tangs of the wood saw and hacksaw blades and the scraper, as applicable. The average of the three determinations of the blade and the tang thickness shall be reported.

4.5.6 Rockwell hardness. The Rockwell hardness shall be determined in accordance with FED-STD-151, Method 243, Hardness Scale C. At least three hardness determinations shall be made on the scraper, 3.2.2 and on the blades, at each different hardness location specified in 3.2.1 through 3.2.1.3, as applicable. The hardness determination per tool shall not be averaged.

* 4.5.7 Ax cutting tests. The ax shall be subjected to practical cutting tests of such duration as may be considered reasonable such as cutting through vines, shrubbery, tree sprouts, saplings, etc., to determine the suitability for performing the tasks outlined in the approved instruction booklet.

4.5.8 Saw cutting

4.5.8.1 Wood saw blade. The wood saw blade shall be inserted into the saw and scraper grip and attached. The saw and scraper grip, with the wood saw blade installed, shall be securely attached to an air motor. The attachment shall permit the saw and grip assembly to pivot in the vertical plane. A one pound weight shall be suspended from the end of the wood saw blade and another one pound weight shall be attached to the saw and scraper as in Figure 25. A length of 4 by 3 inch construction grade oak, with the 4 inch tangential surface uppermost, shall be placed in the cutting position as in Figure 25. The length of the motor stroke shall be adjusted so that the full length of the blade shall be engaged at each stroke. The cycling rate shall be adjusted for 42 $\pm$ 2 cycles per minute. Figure 25 is an example of the wood saw blade cutting inspection apparatus. The oak shall be cut for a depth of one inch and the time recorded from the instant the blade starts cutting until a depth of one inch has been achieved. Four cuts in different places in the oak shall be made and the time required to reach a depth of one inch each cut shall not exceed the time specified in 3.7.2.1. Upon completion of the four cuts, the blade shall be examined for conformance to 3.7.2.1.

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4.5.8.2 Hacksaw blade. The assembled hacksaw blade shall be inserted into the saw and scraper grip and attached. The assembly shall be hand operated and shall completely cut through a piece of cold rolled steel tubing, 3/8 inch in diameter, with the wall thickness 1/16 inch. The cutting rate shall be 60 ± 2 cycles per minute. Eight sections shall be cut from tubing and the assembled hacksaw blade and the saw and scraper grip shall be examined for conformance to 3.7.2.2.

4.5.9 Stress of the scraper. The scraper shall be secured to the upper and lower jaws of a suitable inspection apparatus in such a manner that the applied force tends to straighten the right angle between the blade and tang. The rate of jaw separation, under no load, shall be 12 ± 1/2 inches per minute. A 500 pound load shall be applied and held for 30 seconds. After 30 seconds, the 500 pound load shall be released. The scraper shall then be examined for conformance to 3.7.3.

4.5.10 Sharpening of the ax blade. The sharpening stone shall be used to sharpen and hone the cutting edges of the ax blade. Each side of the sharpening stone shall be used. The sharpening stone shall than be inspected for conformance to 3.7.4.

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. The sharpening stone shall be placed within its carrying case and the case shall be snapped closed. The sharpening stone and the fire starter shall be placed in separate pockets of the carrying case of the type I or IV kit, as applicable. The leg strap shall be neatly folded and shall be placed within the pocket of the ax carrying case containing the fire starter. The ax shall be placed within its carrying case and the neck band shall be snapped closed around the ax of the type I or type IV kit, as applicable. The case shall be closed by the slide fastener. The wood saw and hacksaw blades shall be placed flat in the pocket of the carrying case of the type I kit. The saw and the scraper grip and scraper shall be placed between the carrying case and its pocket in a manner that shall not damage the case or the contents. The case shall be closed by the slide fastener. If the type I kit is being purchased, the male snap fasteners on the underside ax carrying case shall be mated to the female snap fasteners on the upperside saw carrying case and snapped together. The kits of the one type shall be packaged in the most compact manner with a snug-fitting fiberboard container conforming to PPP-B-636, type CF, variety SW, grade 200. The weight of the contents of each package shall not exceed 30 pounds. The containers shall be constructed and closed in accordance with the appendix to PPP-B-636. All the seams and joints shall be sealed with a minimum of 3 inch wide tape conforming to PPP-T-45, type III, grade A, B, or C.

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5.1.2 Level C. The kits of one type shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2). Shipping containers, insofar as possible, shall be uniform in size and shape and of minimum cube and tare weight. Each shipping container shall only contain kits of one type.

5.2.1 Level A. The kits, packaged as specified in 5.1, shall be packed as specified in 5.2.2, except that the fiberboard container shall be type SF, weather resistant class, variety SW, grade V2s. Each container shall be constructed, closed, waterproofed, and reinforced with flat strapping or tape banding in accordance with the appendix to PPP-B-636.

5.2.2 Level B. The kits, packaged as specified in 5.1, shall be packed with a snug-fitting fiberboard shipping container conforming to PPP-B-636, type CF or SF, domestic class, variety SW, grade 275. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636, method II. The weight of the contents within the shipping container shall not exceed 65 pounds.

5.2.3 Level C. The packaged kits, that require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that shall insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special marking required by the contract or order, the interior packages and the exterior containers shall be marked in accordance with MIL-STD-129 and shall include the type of the kit.

6. NOTES

6.1 Intended use. The kits covered by this specification are intended for use by airmen as an aid to subsistence upon forced landings in a remote and unfavorable, hostile environment.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification

b. Type and quantity desired (see 1.2)

c. Whether first article inspection is waived (see 4.3.1)

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d. Name and address of the first article inspection laboratory (see 4.3.1)

e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

f. Whether any special markings are required (see 5.3)

6.3 Data. For the information of the Contractors and Contracting Officers, any of the data specified in applicable documents listed in section 2 of this specification or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.4 Changes from previous issue. Asterisks are used in this revision to identify changes with resect to the previous issue.

6.5 Supersession data. The type I kit is a combination of the deleted type II kit (ax with wood handle) and the type III kit (saw).

6.6 Subject term (key word) listing.

Survival kits
Individual

Custodians:
Army - GL
Navy - AS

Review activities:
Army - GL
Navy - AS
Army - AV
Navy - MC

Preparing activity:
DLA - CT

(Project No. 8465-0127)

NOTICE - Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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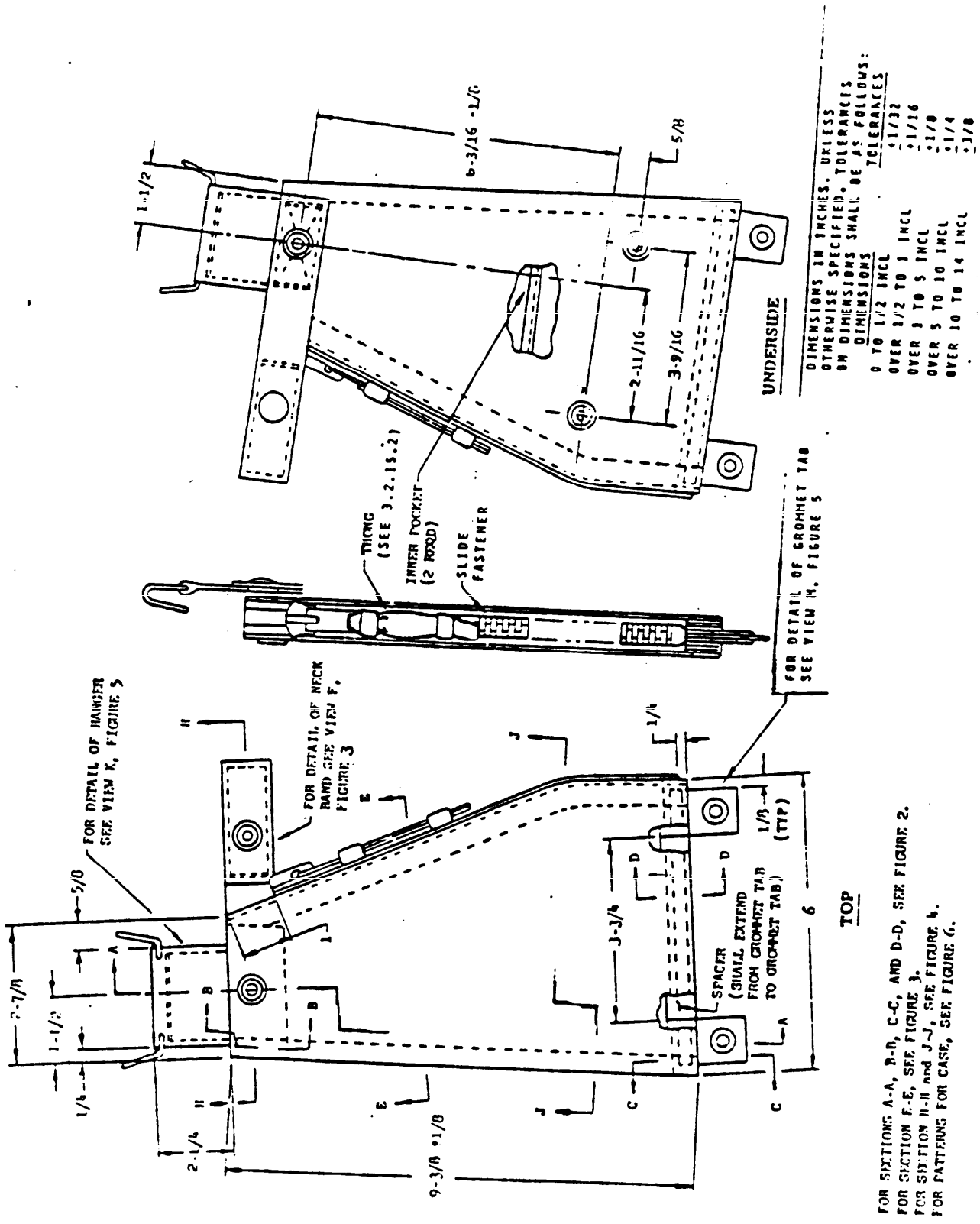
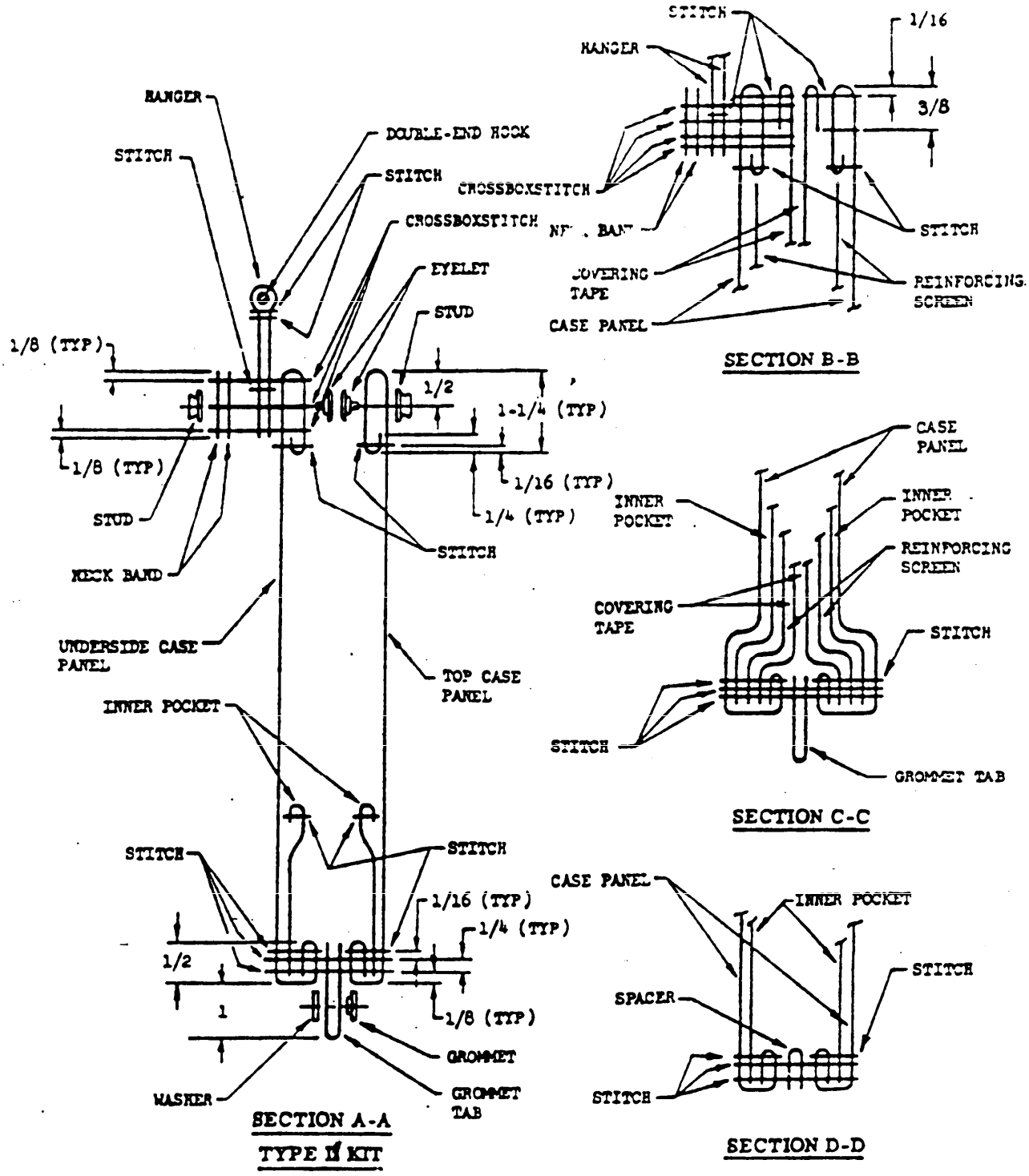


FIGURE 1. Ax carrying case assembly for type I kit.

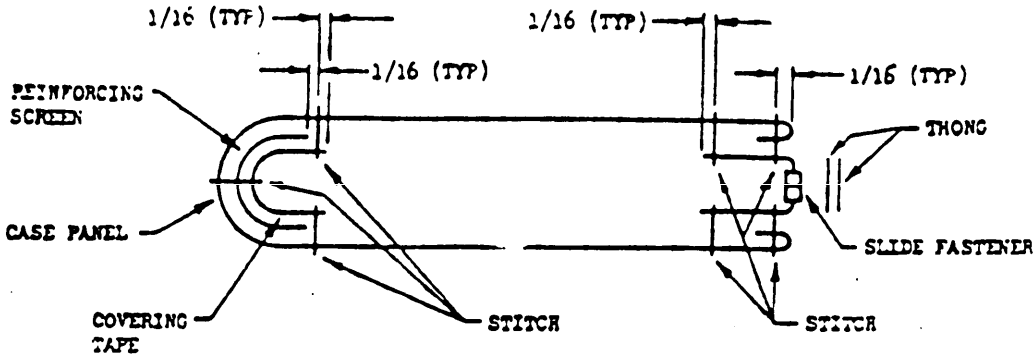
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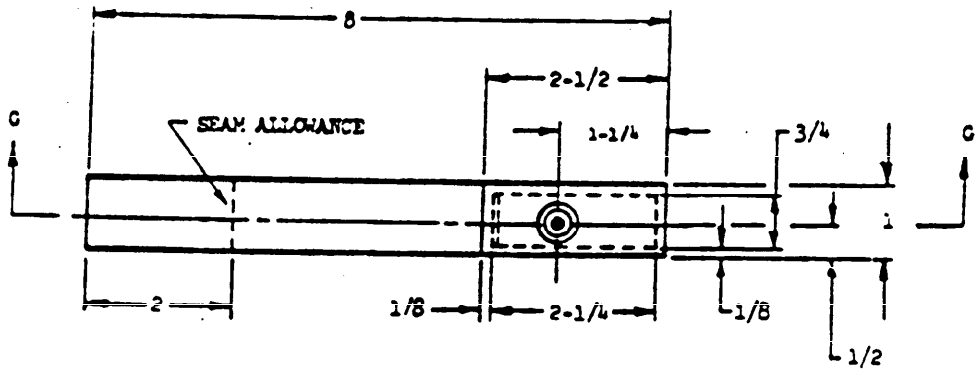
DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

FIGURE 2. Ax carrying case sections for type I and IV kits.

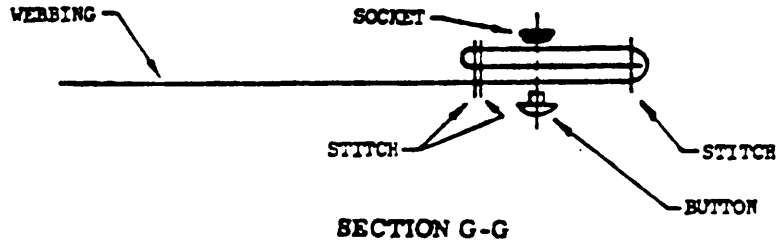
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SECTION E-E



VIEW F
NECK BAND ASSEMBLY
TYPE I KIT

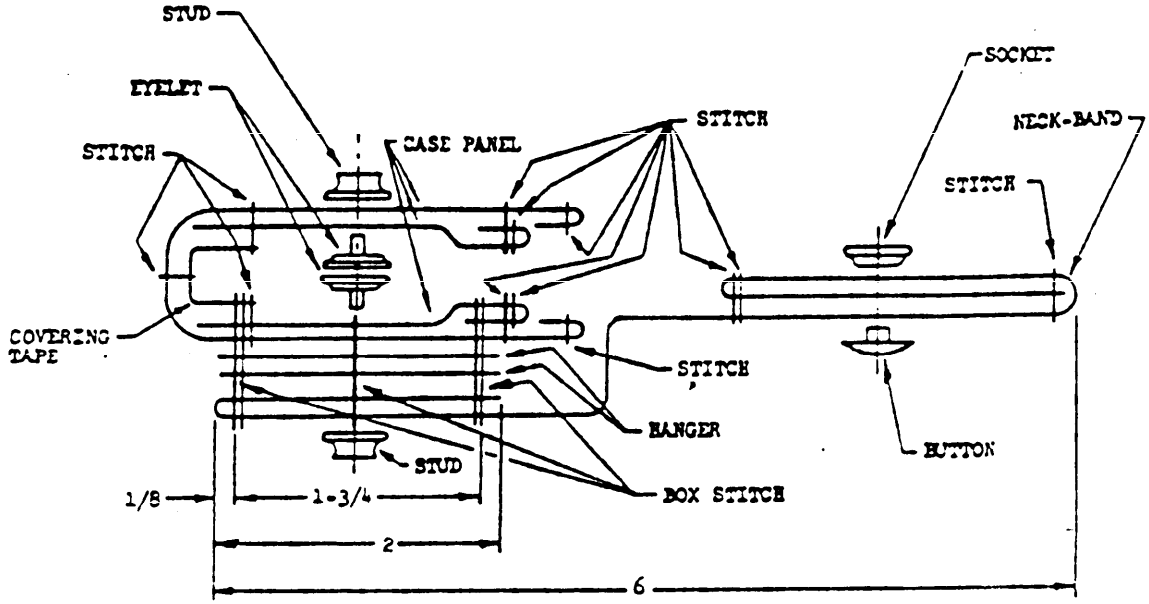


SECTION G-G
TYPE I KIT

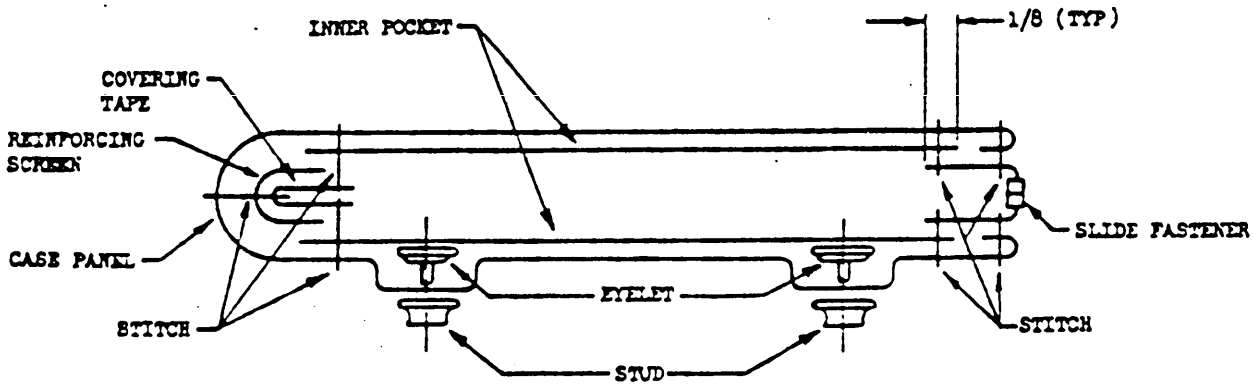
DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

FIGURE 3. Ax carrying case sections and neck and assembly for types I and IV kits.

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SECTION H-H

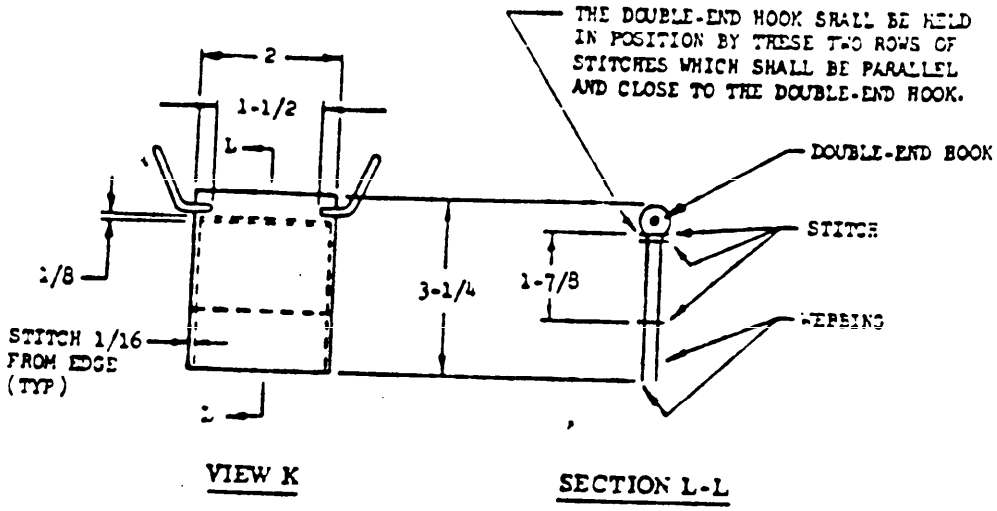


SECTION J-J

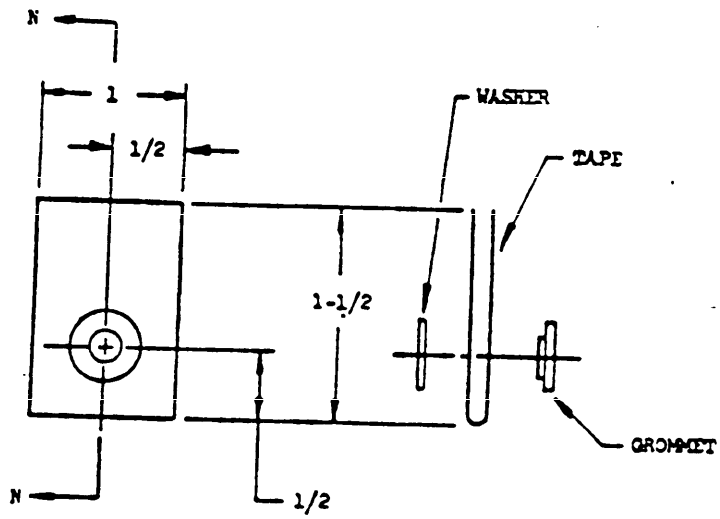
DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

FIGURE 4. Ax carrying case sections for type I kit.

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HANGER ASSEMBLY
TYPE II KIT

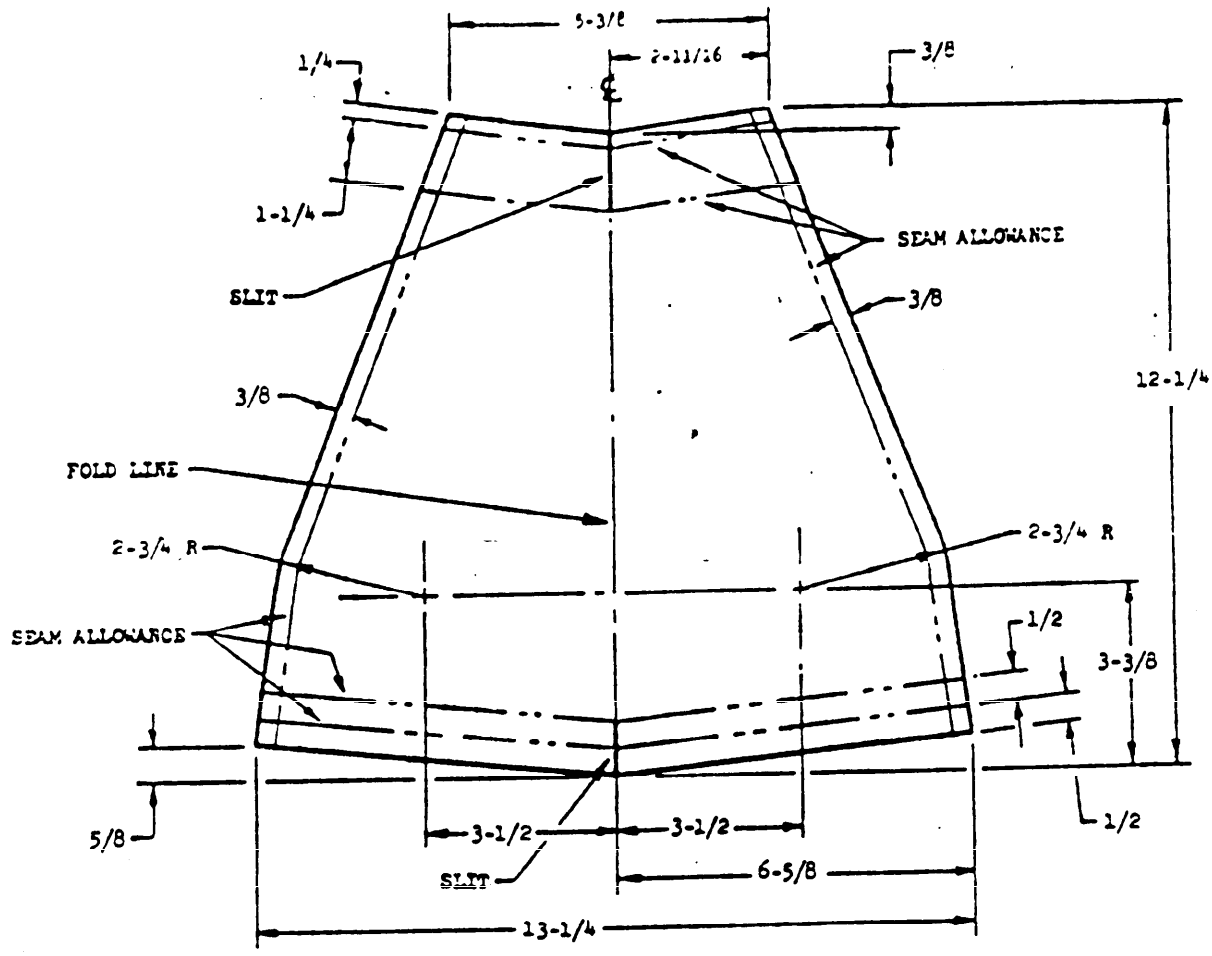


GROMMET TAB ASSEMBLY

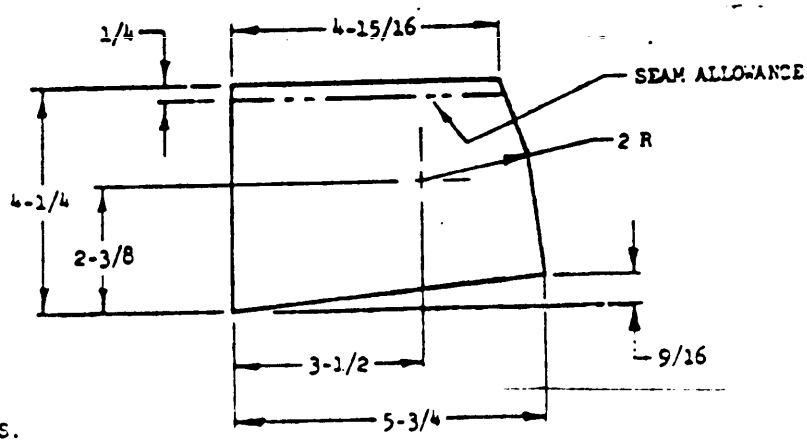
DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

FIGURE 5. Hanger and grommet tab assemblies for types I and IV kits

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CASE PANEL



DIMENSIONS IN INCHES.
THE DIMENSIONS OF THE ASSEMBLED CARRYING CASE SHALL CONFORM TO FIGURES 1 THROUGH 5.

POCKET PANEL

FIGURE 6. Flat patterns for type I kit ax carrying case.

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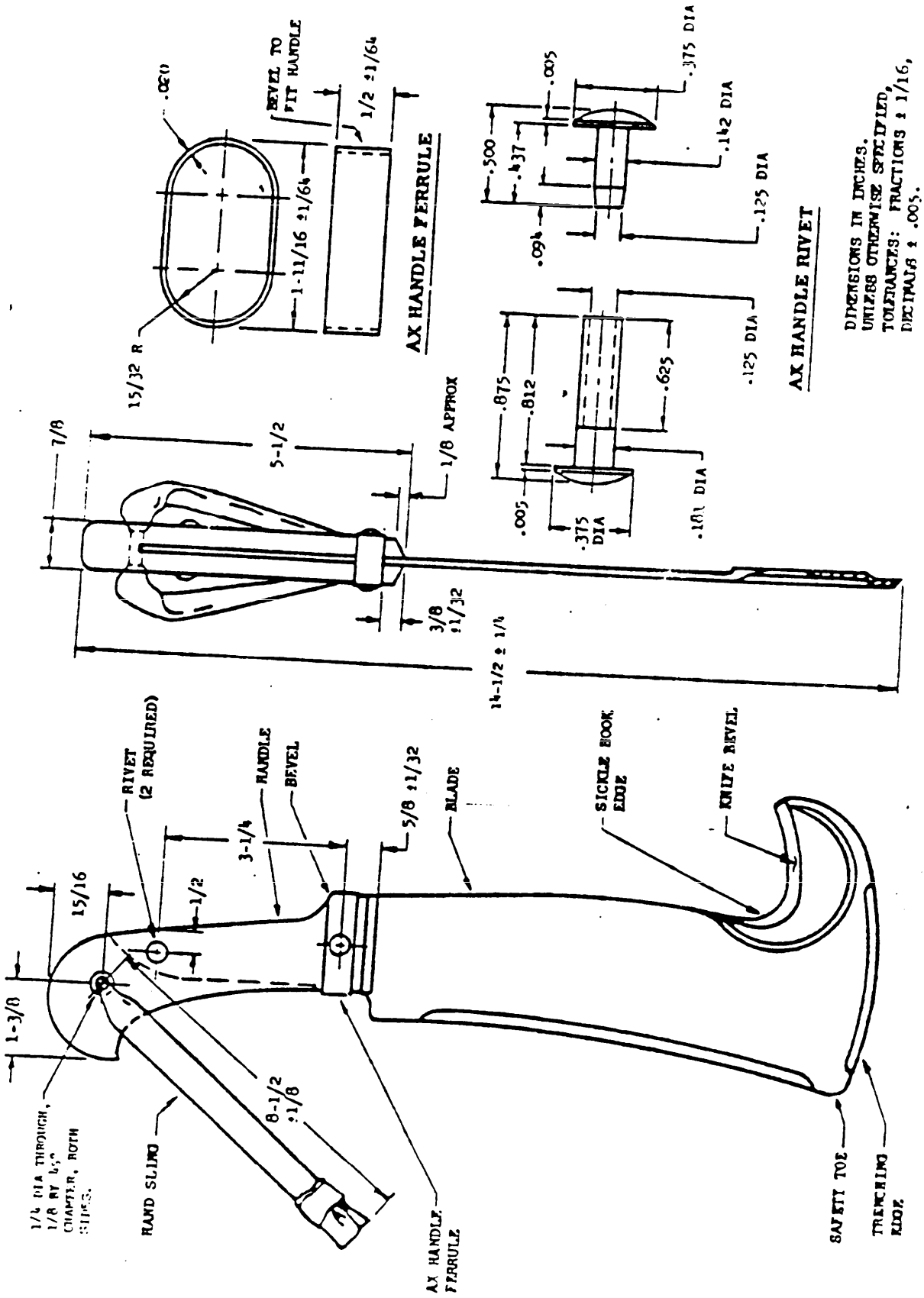


FIGURE 7. Ax assembly and details for type I kit.

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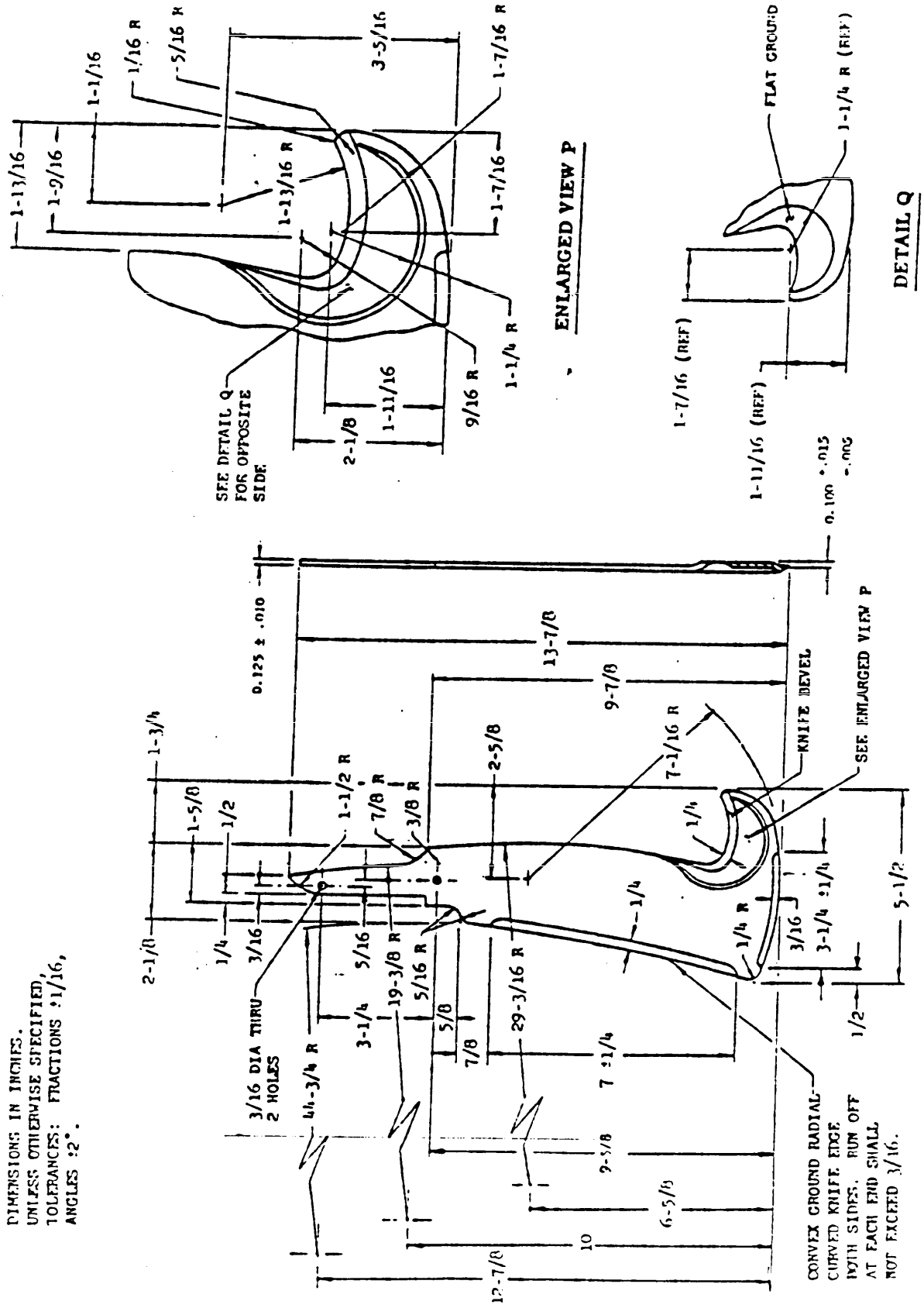
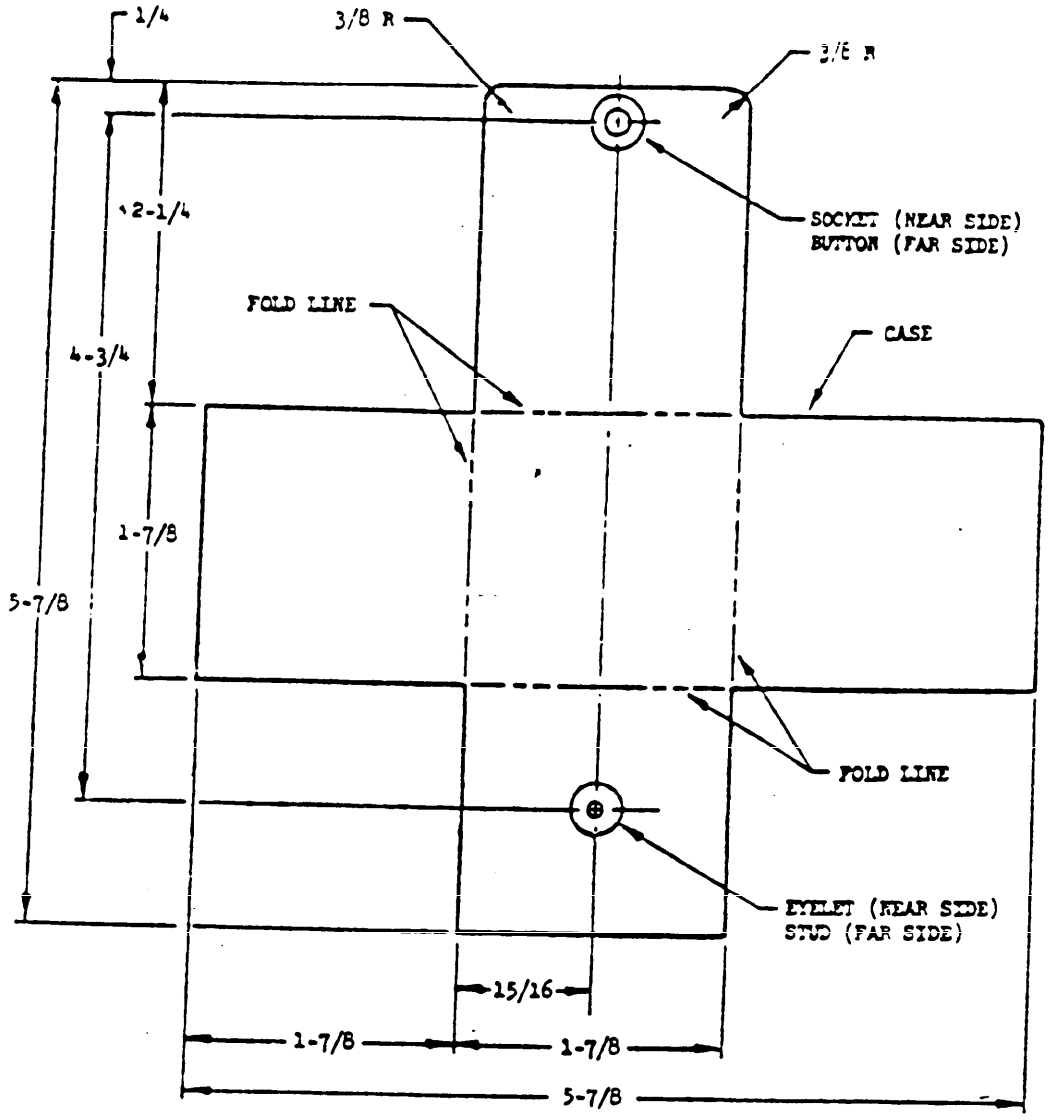


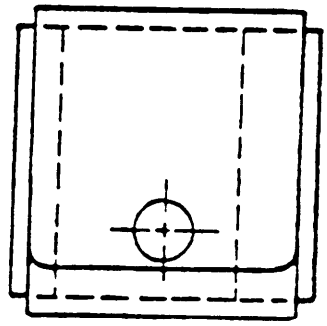
FIGURE 8. Ax blade details for type I kit.

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SHARPENING STONE CASE (OPENED)

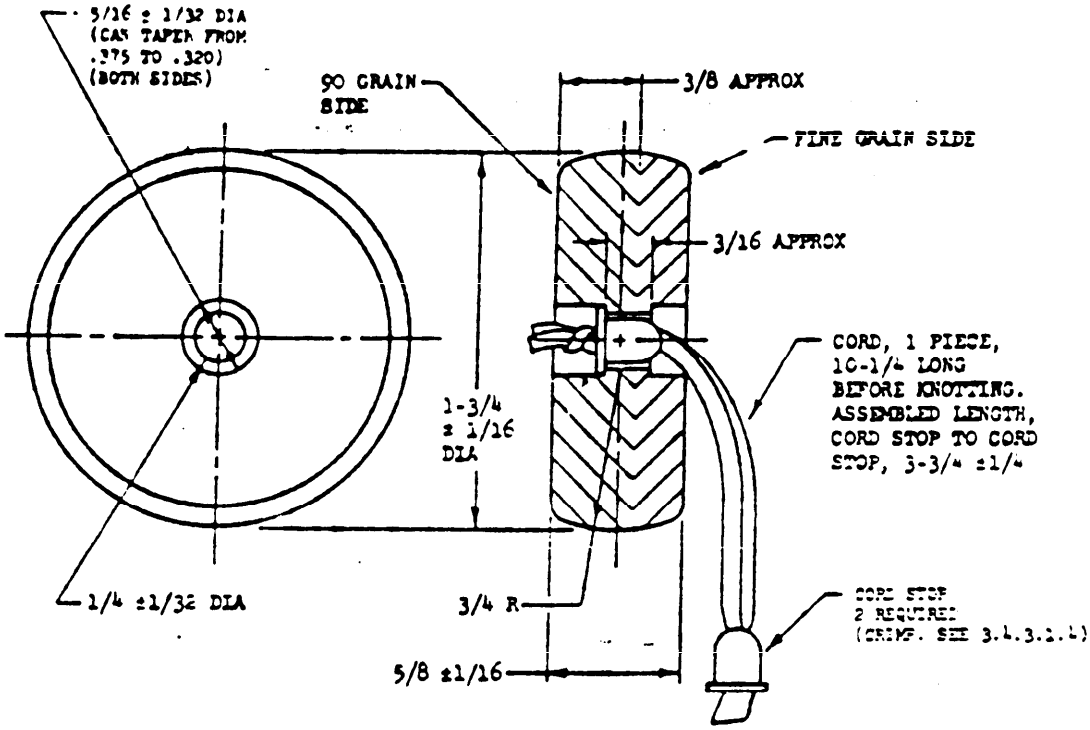
DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.



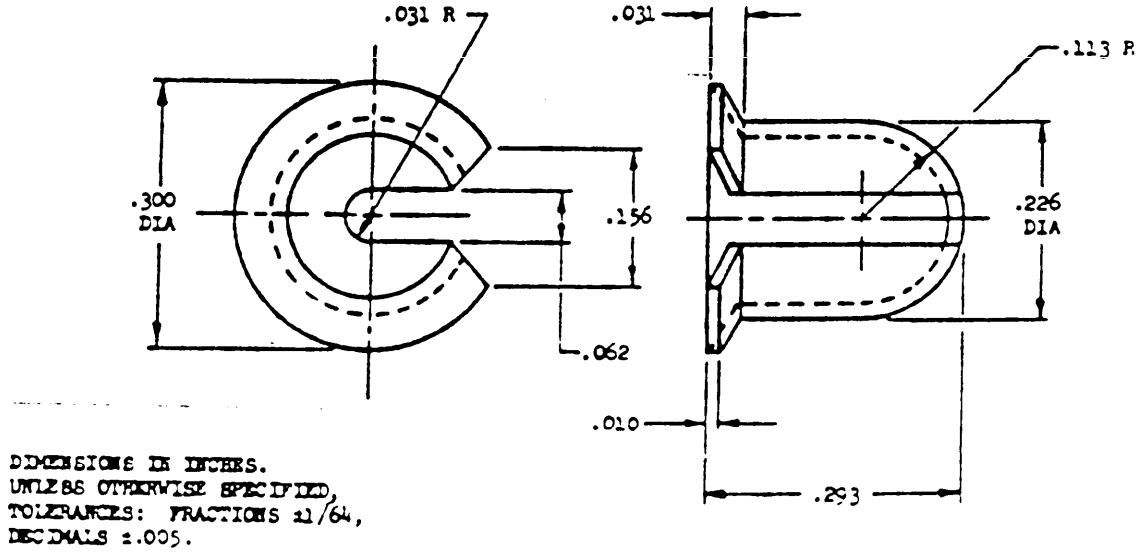
SHARPENING STONE CASE (CLOSED)

FIGURE 9. Sharpening stone case for types I and IV kits.

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SHARPENING STONE ASSEMBLY
(ONE PIECE CONSTRUCTION)



CORD STOP

FIGURE 10. Sharpening stone assembly and details for types I and IV kits.

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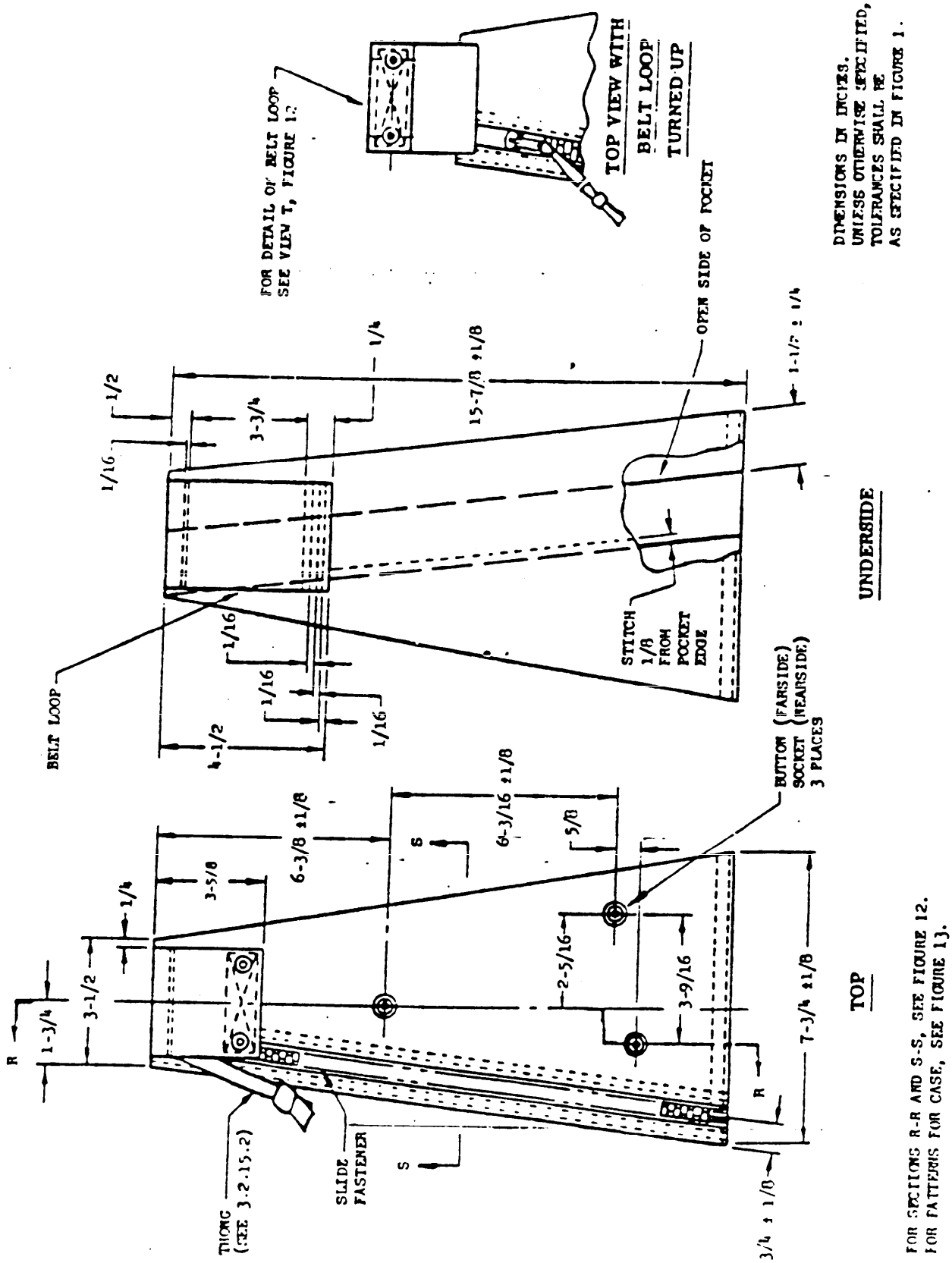


FIGURE 11. Ax carrying case assembly for type I kit.

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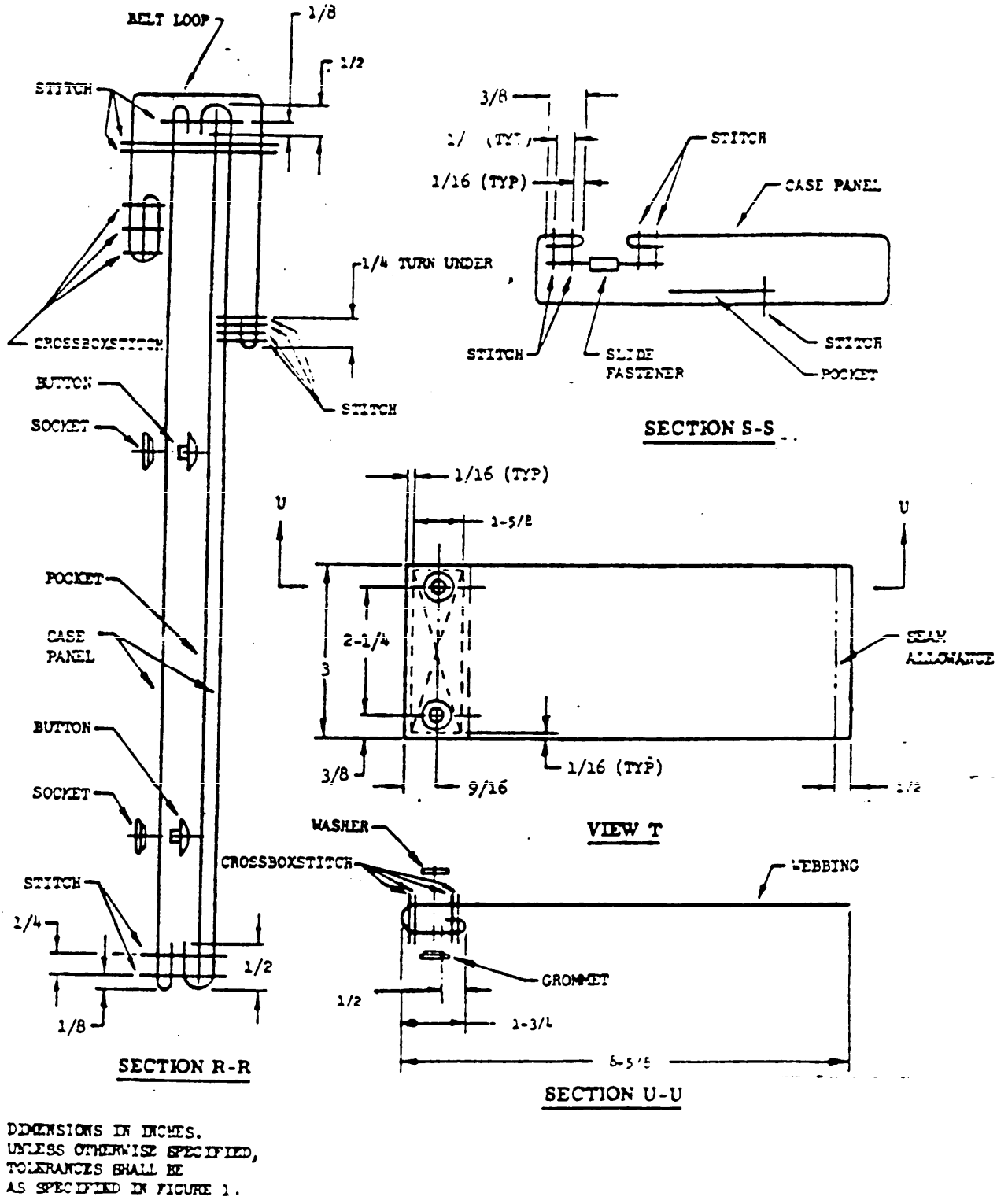
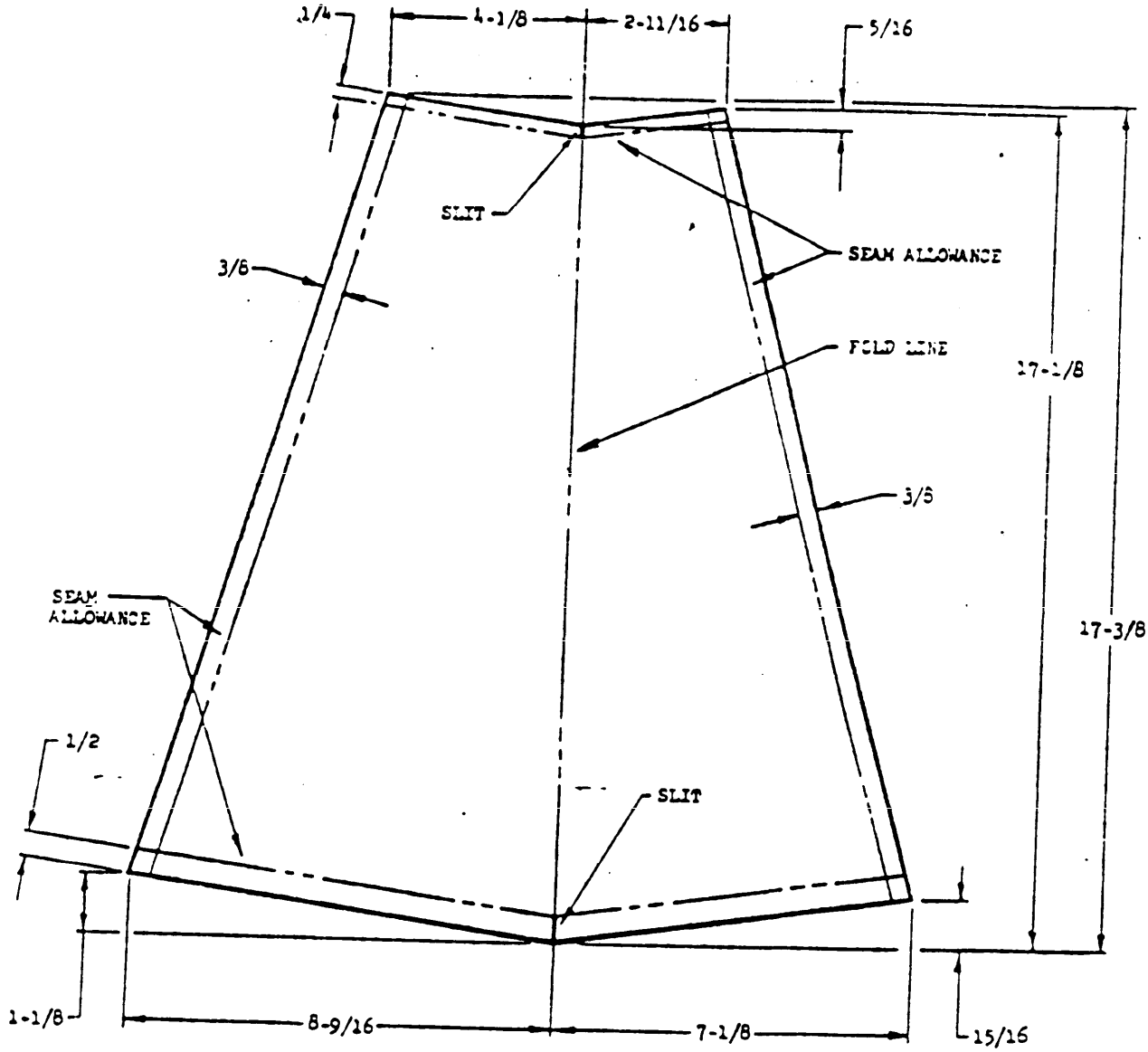


FIGURE 12. Ax carrying case sections for type I kit.

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Rosin Acid



CASE PANEL

DIMENSIONS IN INCHES.
THE DIMENSIONS OF THE ASSEMBLED CARRYING CASE SHALL CONFORM TO FIGURES 11 AND 12.

FIGURE 13. Flat pattern for type I kit saw carrying case.

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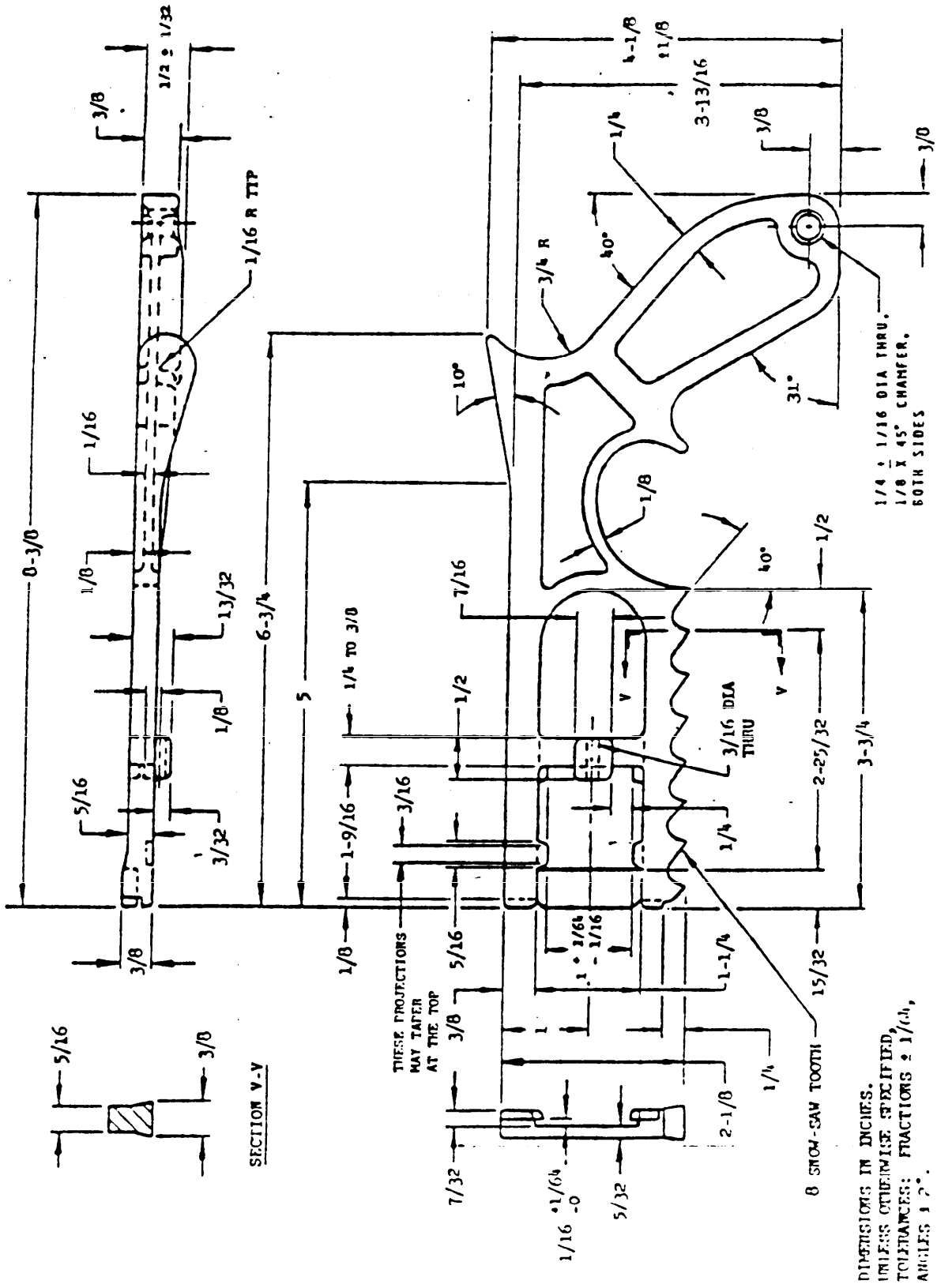
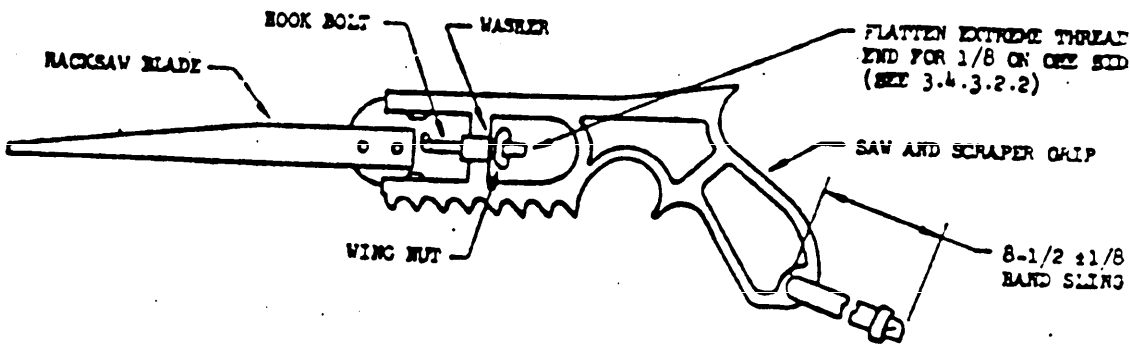
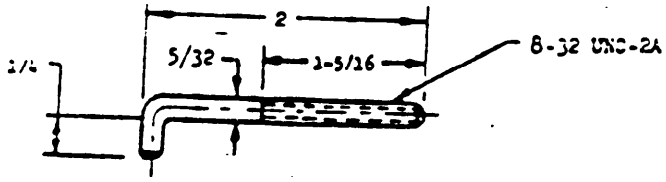


FIGURE 14. Saw and scraper grip details for type I kit.

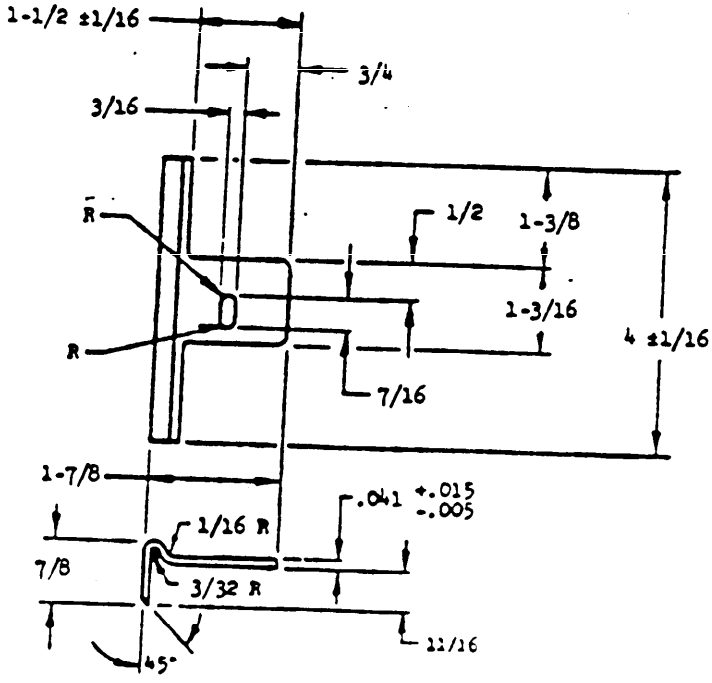
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SAW AND SCRAPER GRIP ASSEMBLY WITH HACKSAW BLADE ATTACHED
(SEE 6.3)



HOOK BOLT

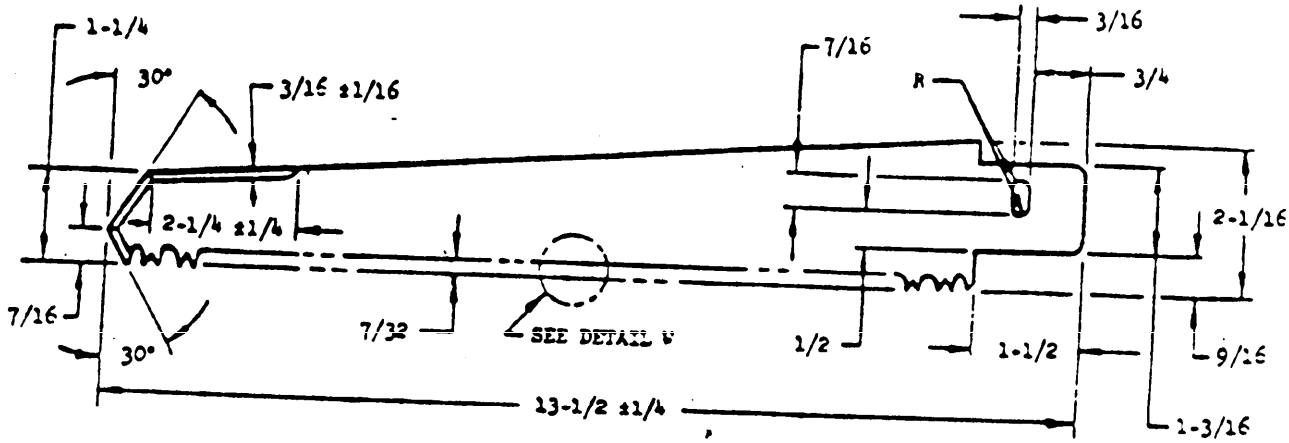


DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64,
ANGLES ± 2°.

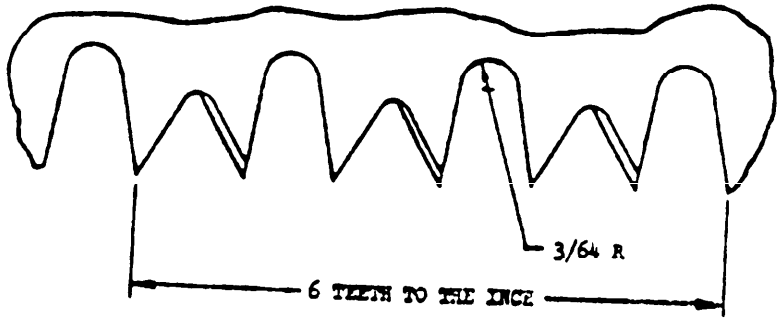
SCRAPER

FIGURE 15. Saw and scraper grip details and assembly and scraper details for type I kit.

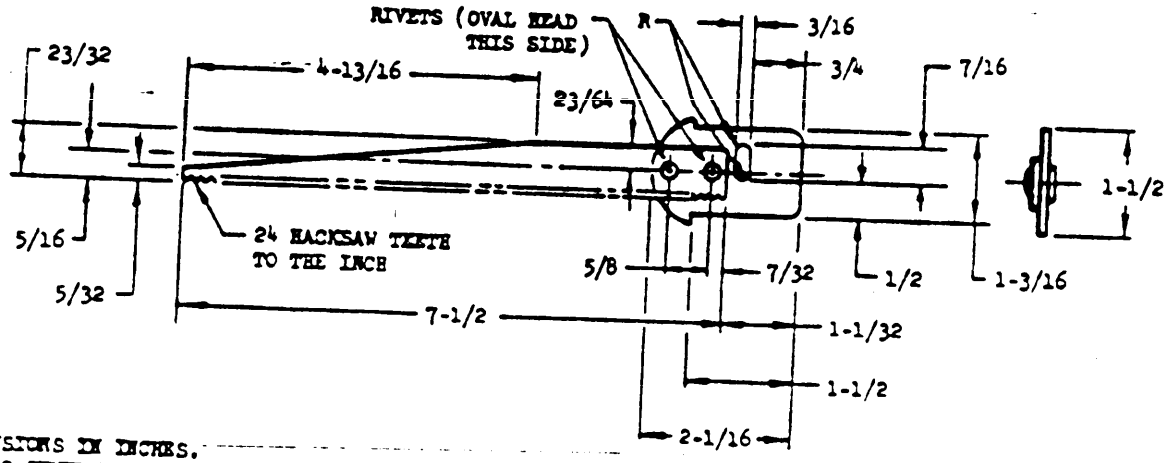
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WOOD SAW BLADE



DETAIL W



DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES: FRACTIONS ± 1/64,
ANGLES ± 2°.

HACKSAW BLADE

(SEE 6.3)

FIGURE 16. Wood saw and hacksaw blades details for type I kit.

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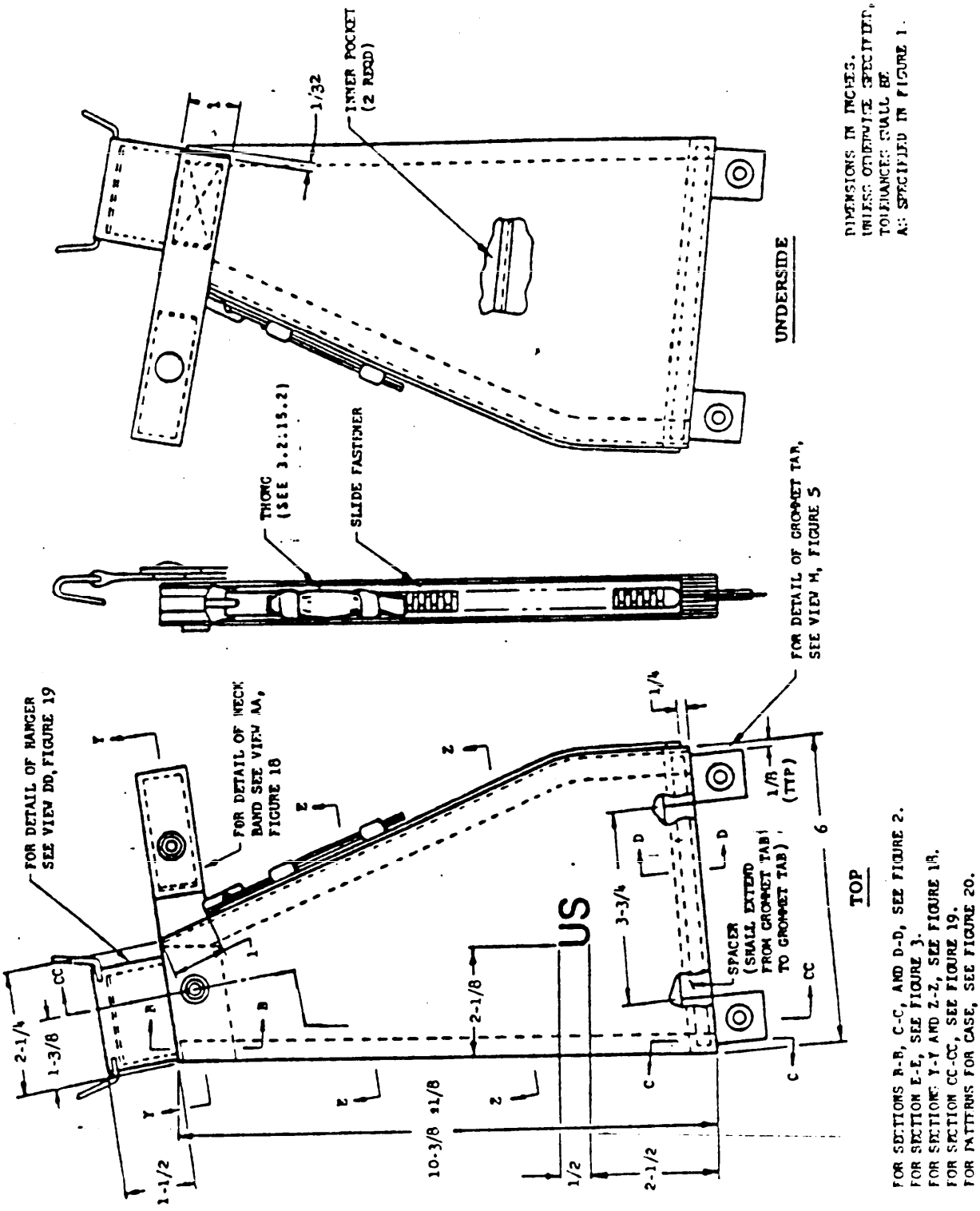


FIGURE 17. Ax carrying case assembly for type IV kit.

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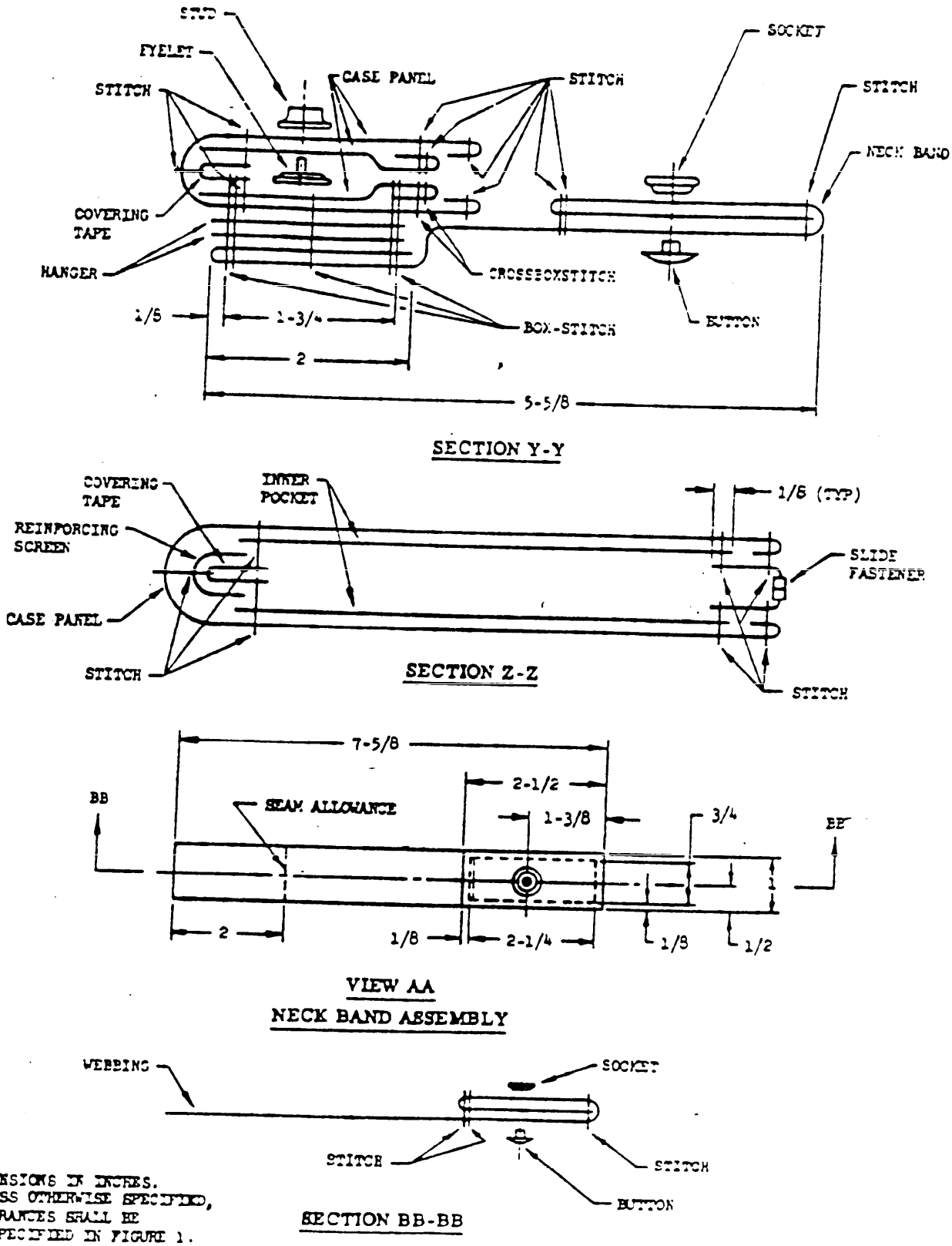
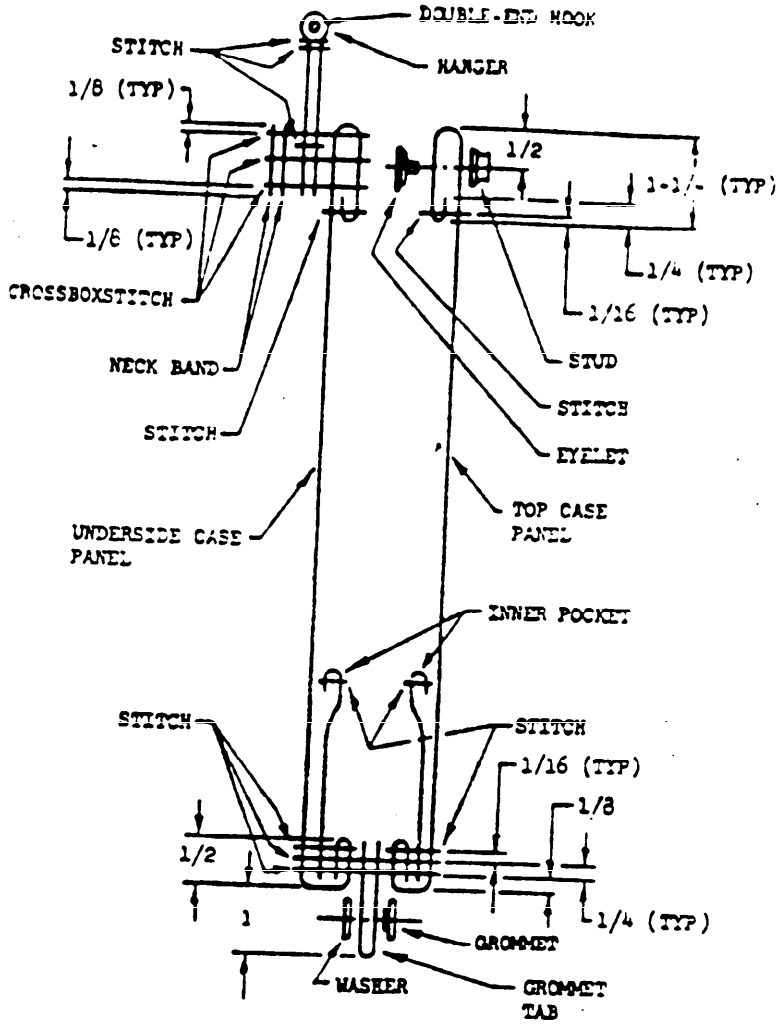
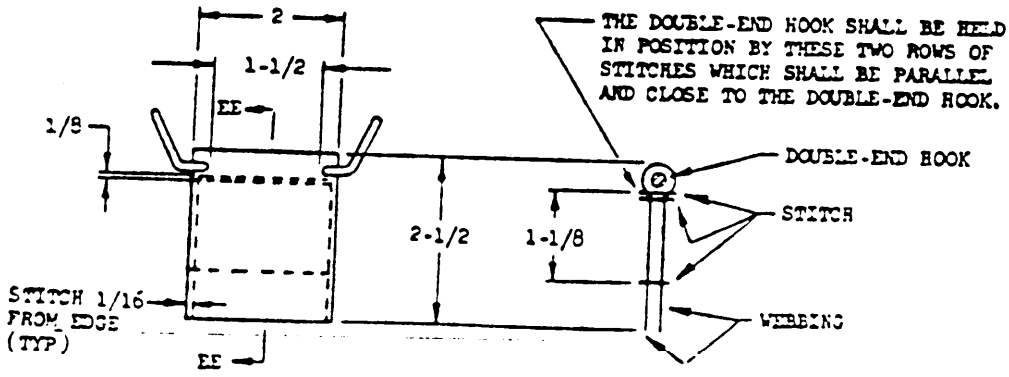


FIGURE 18. Ax carrying case sections for type IV kit.

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SECTION CC-CC



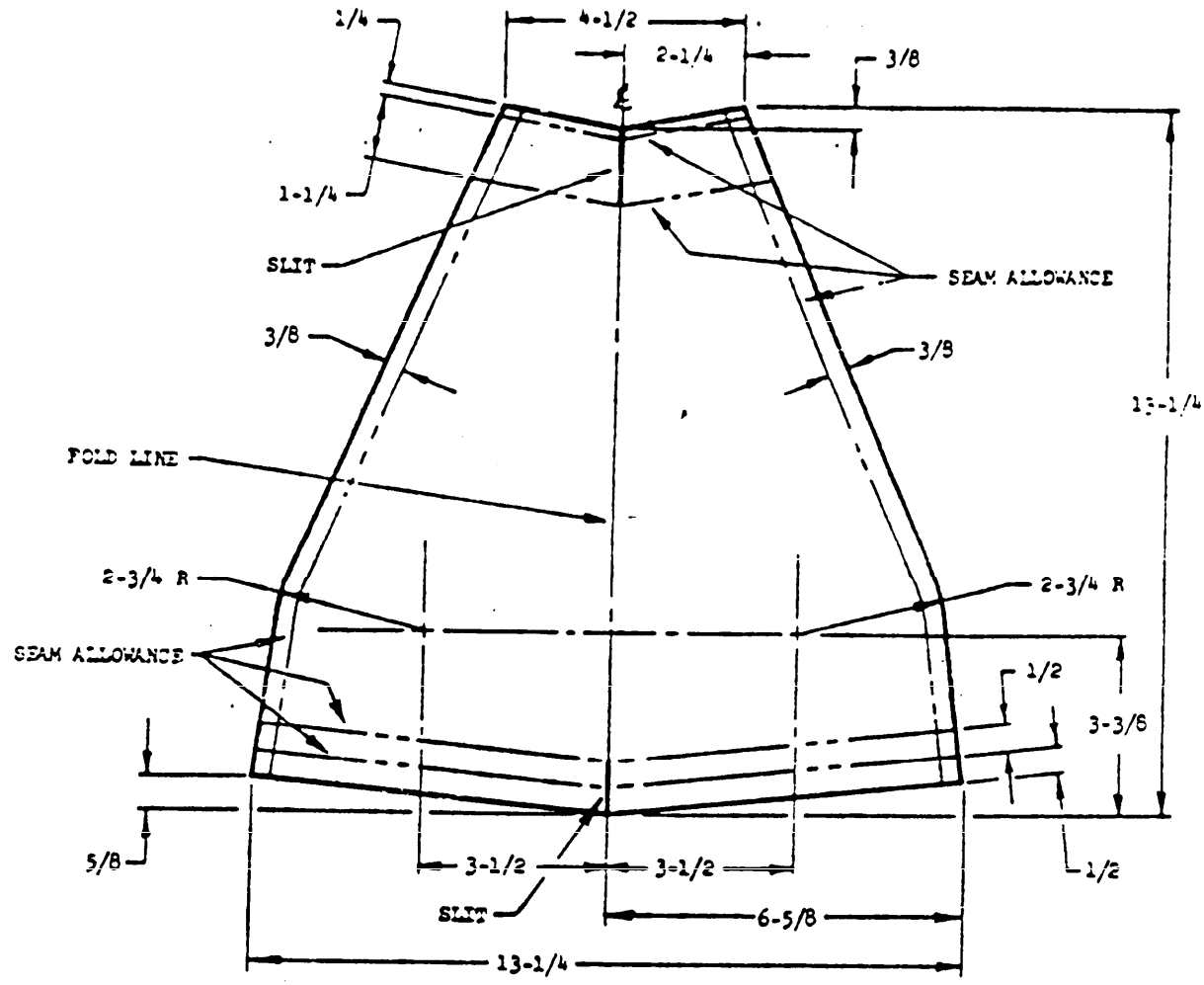
DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE AS SPECIFIED IN FIGURE 1.

VIEW DD

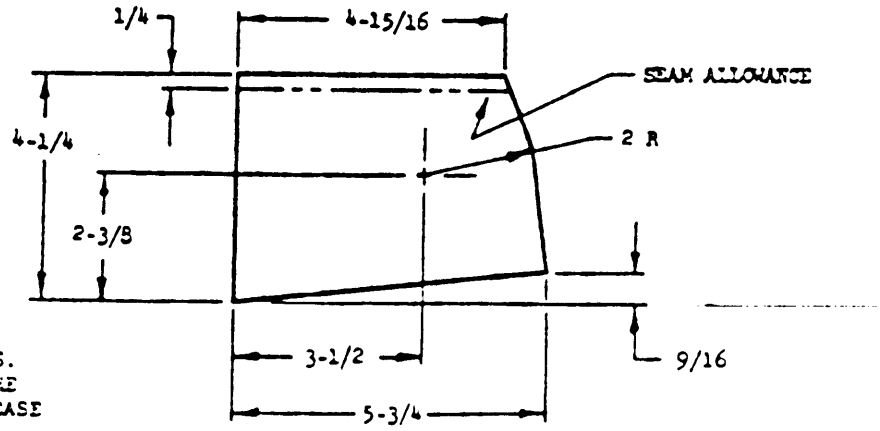
SECTION EE-EE

FIGURE 19. Ax carrying case section for type IV kit.

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CASE PANEL



DIMENSIONS IN INCHES.
THE DIMENSIONS OF THE ASSEMBLED CARRYING CASE SHALL CONFORM TO FIGURES 2, 3, 5, AND 17 THROUGH 19, AS APPLICABLE.

POCKET PANEL

FIGURE 20. Flat patterns for type IV kit ax carrying case.

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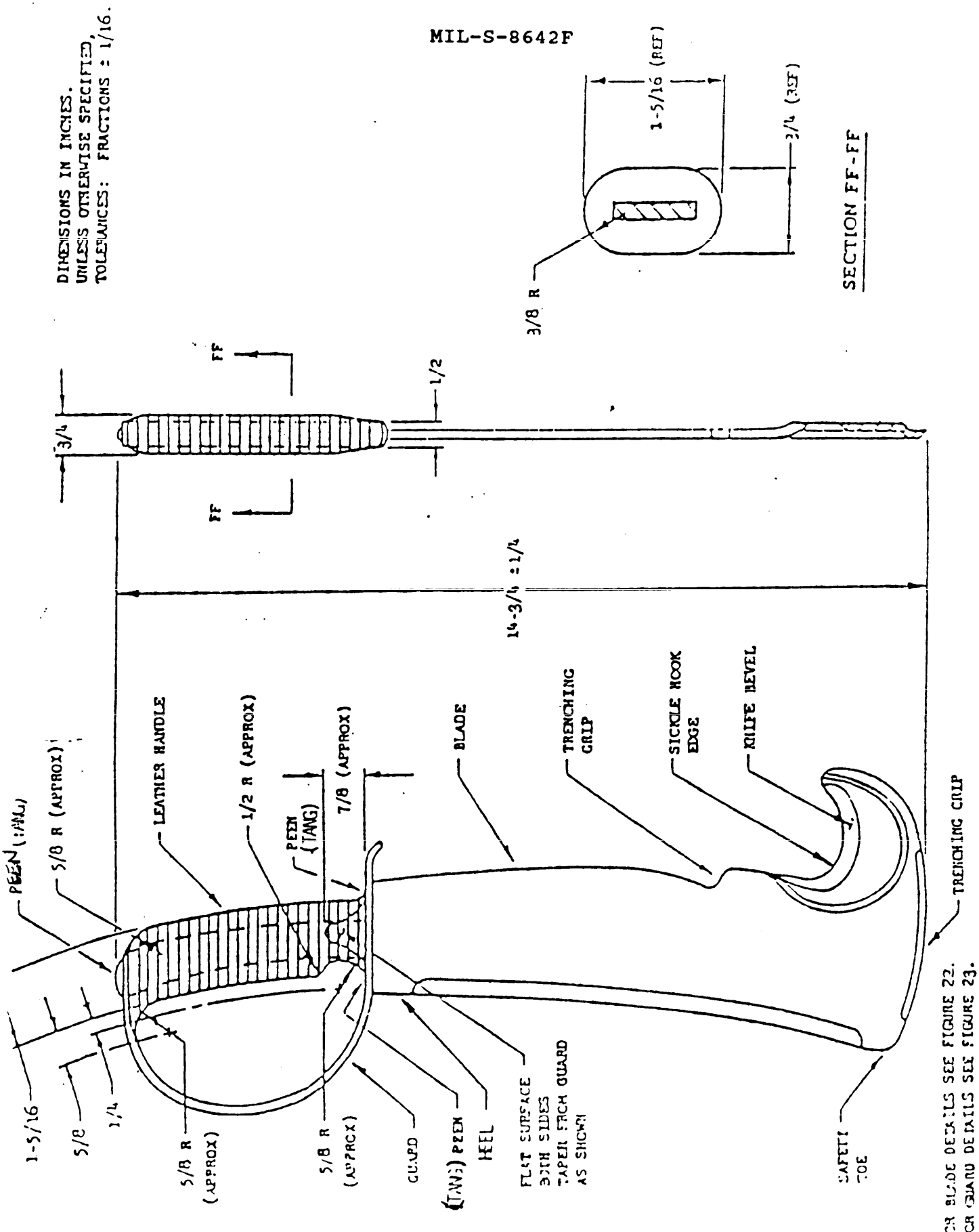


FIGURE 21 Ax assembly for type IV kit.

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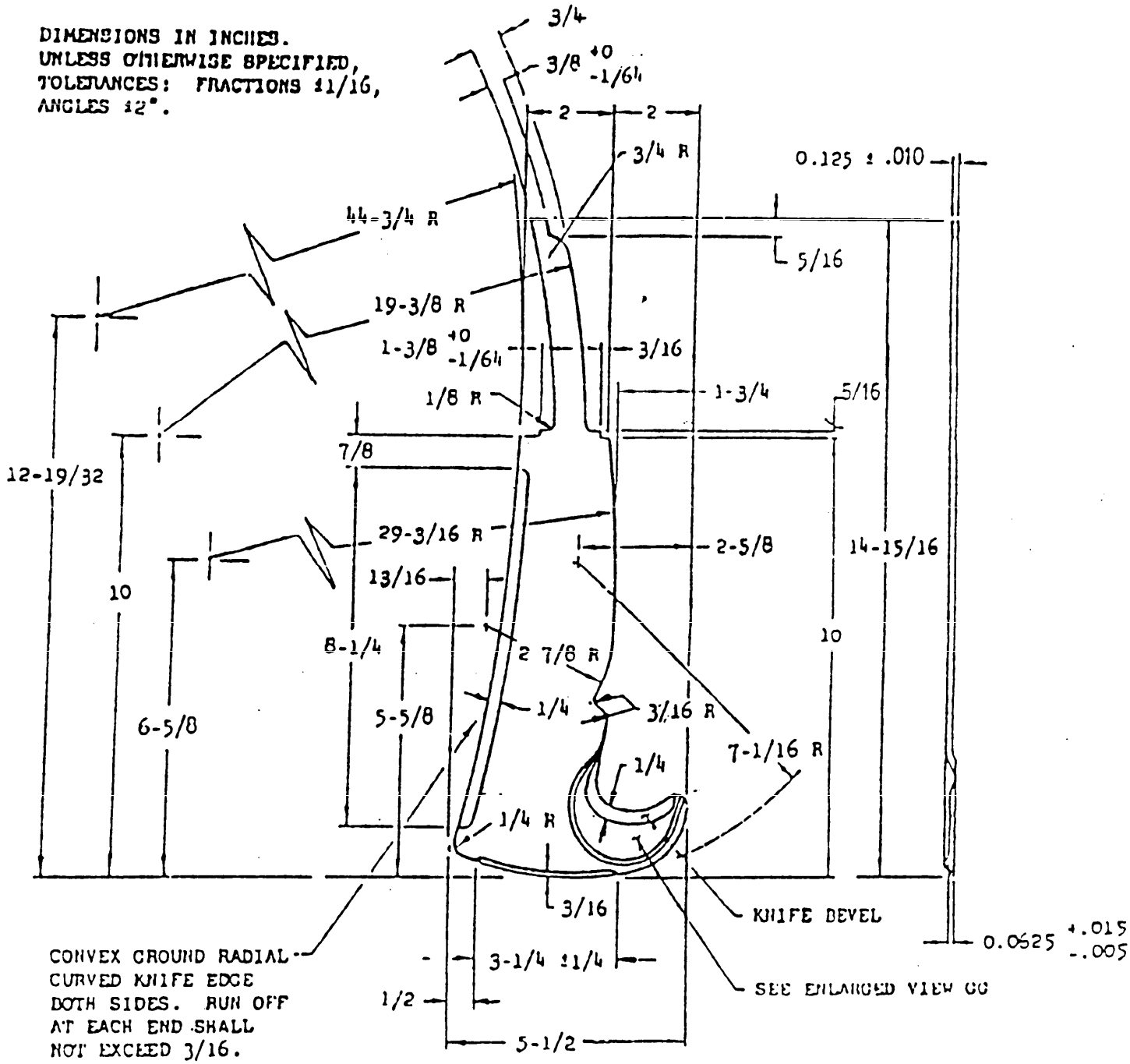


FIGURE 22. Ax blade details for type IV kit.

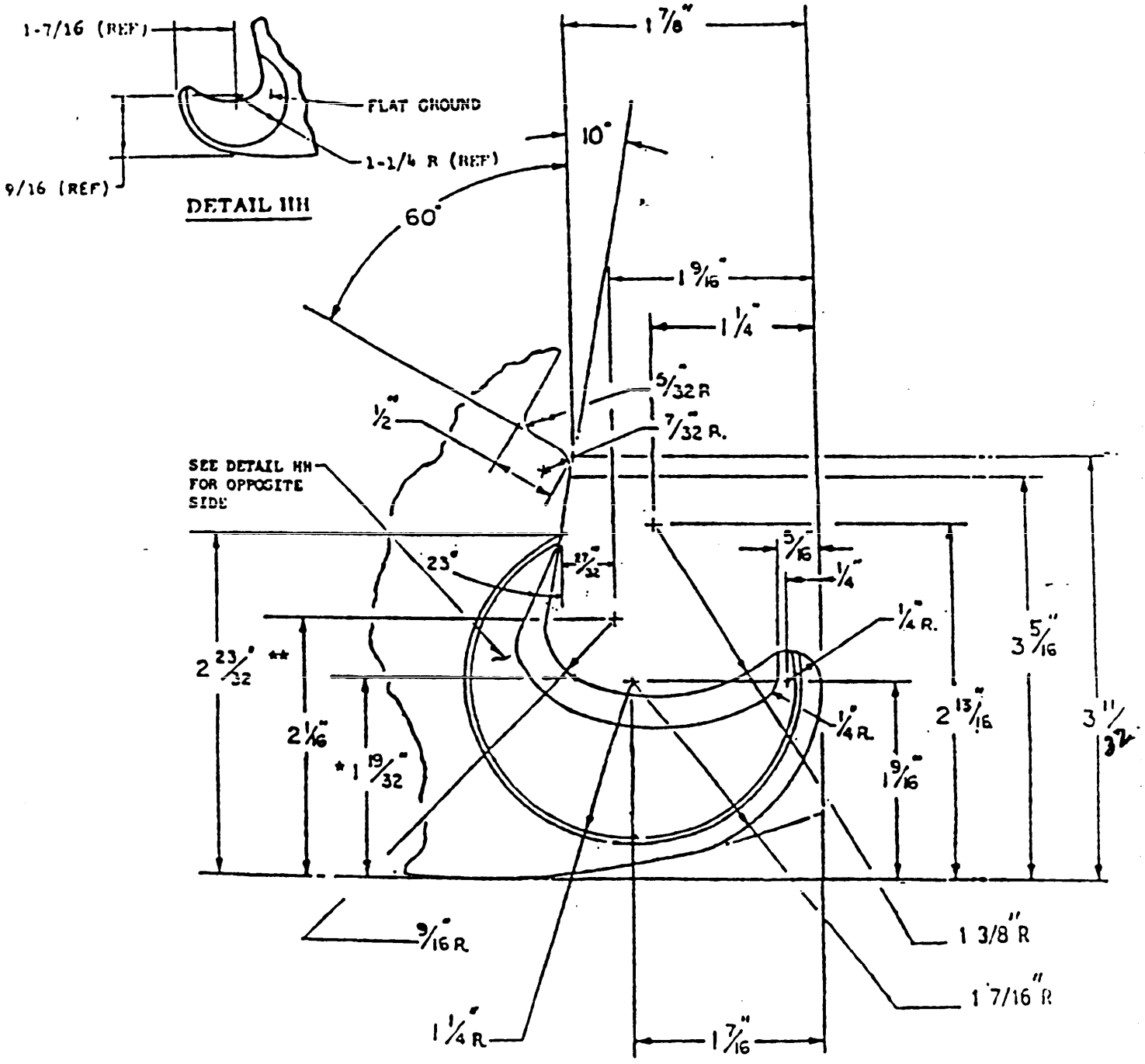
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Note:

1. To form the sickle's tapered knife bevel edge, blend the knife bevel edge from the 9/16 inch radius up to the 23 degree angle starting at the 1 1/4 inch radius relief.

2. * Measurement indicates point where taper begins.

3. ** Measurement indicates where taper ends (meets 23 degree angle).



Enlarged view GG

FIGURE 23. Sickle details for the type IV ax blade

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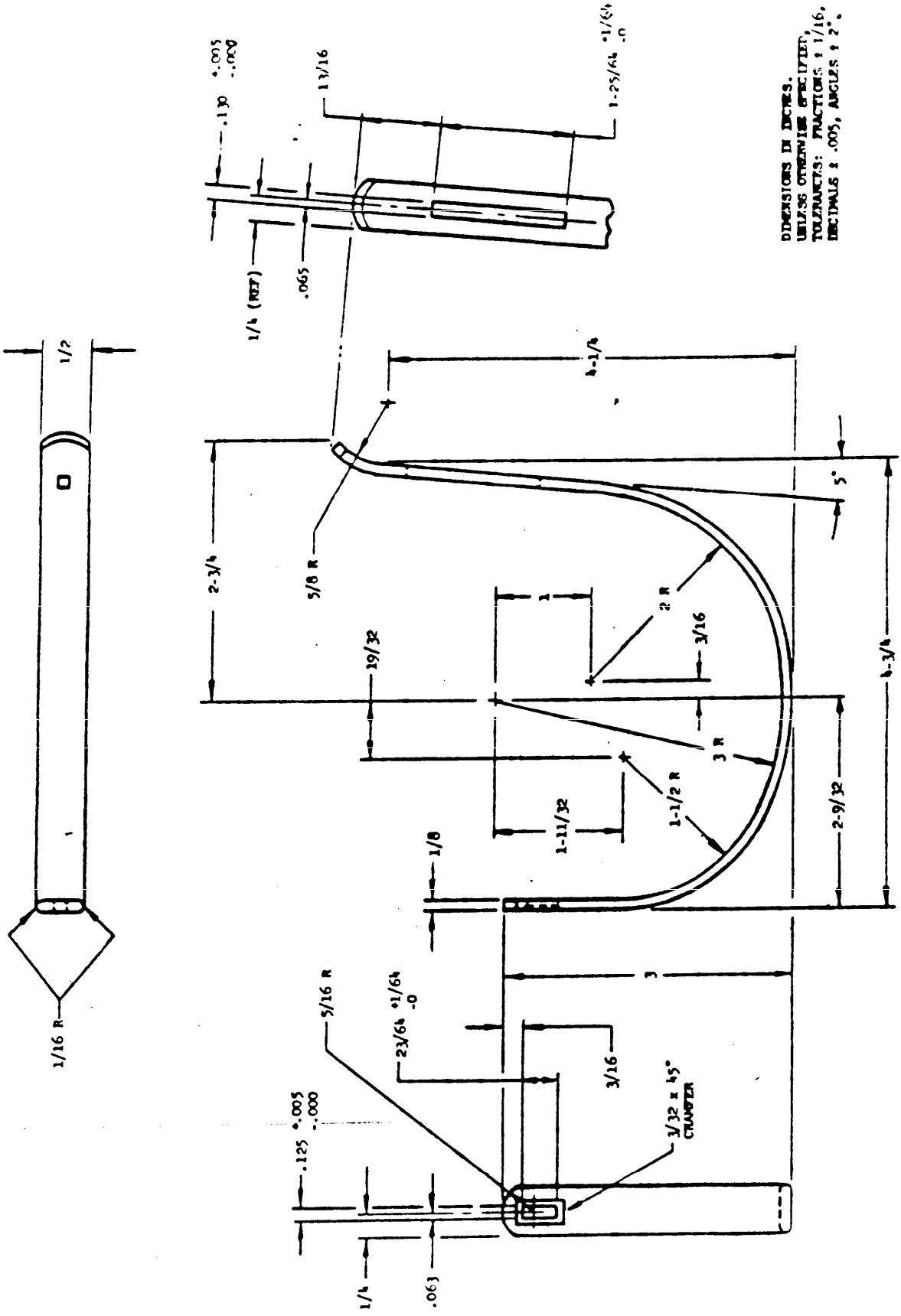


FIGURE 24. Ax guard details for type IV kit.

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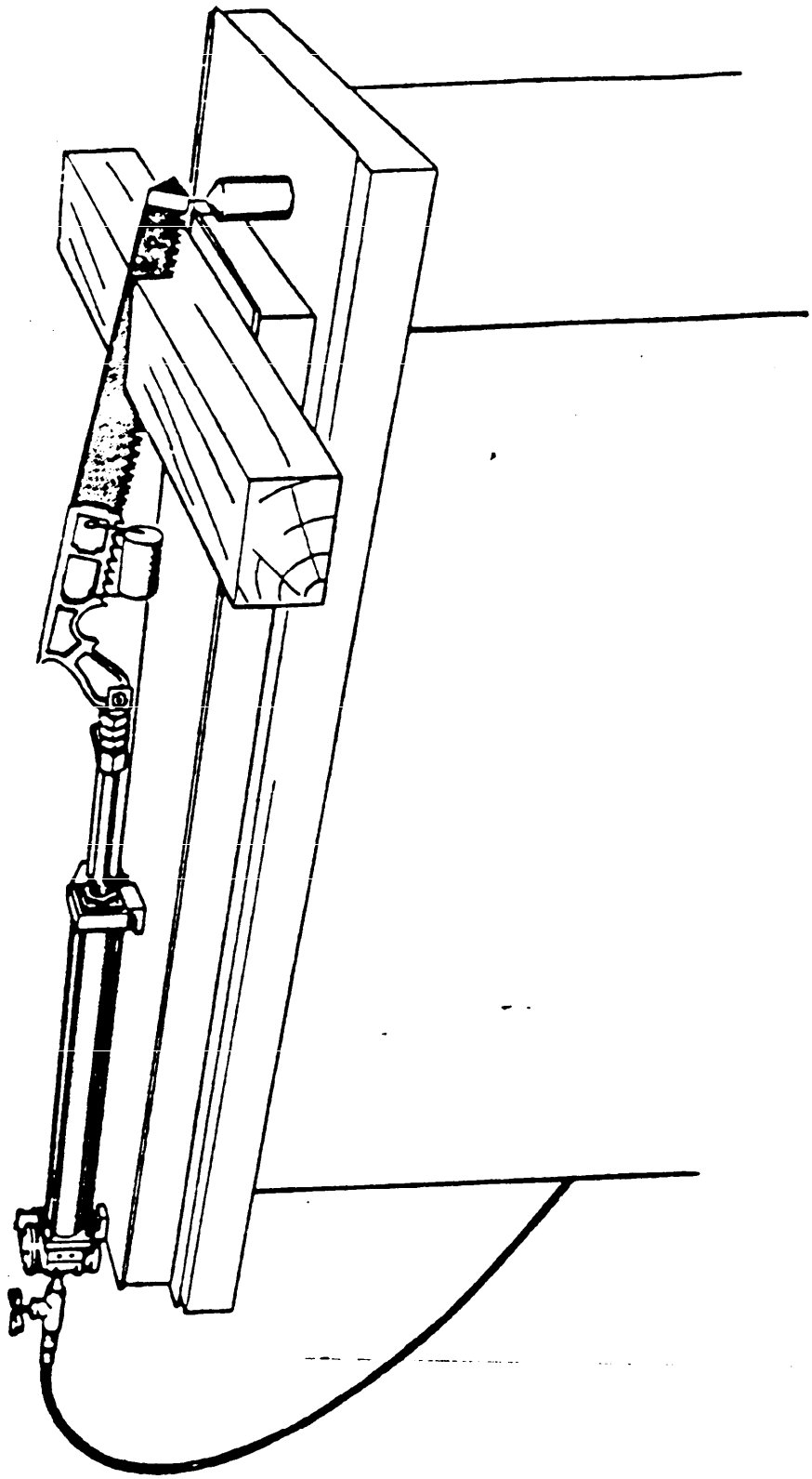


FIGURE 25. Wood saw blade cutting inspection apparatus.

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